

HALO Frontend

Complete Specification — Every Page, Every Element, Every Data Binding

Built from	The full HALO blueprint (all 70 sections, read directly from source text) and the backend's API contract — every route's response read field by field from the code, completed with the fields the backend punch list delivers. Part D is that contract in one place
Why it exists	The blueprint describes what every page must do in detail, but (1) it never maps a page element to the actual API field that feeds it, (2) its functional rules are interleaved sentence-by-sentence with visual values that are now being redesigned, (3) one page's content is scattered across up to eight different sections, and (4) it was written before the backend existed, so it doesn't know the fields the backend actually uses. This document fixes all four.
What's locked	Every page's purpose, content, states, buttons, copy, interactions, and data rules — from the blueprint, unchanged
Design foundation	Locked: colors (brand and severity, light and dark) and typefaces — section A11
What's open	Page-level visual choices — composition, component shapes, spacing, motion values. Listed per page as decisions to make. The frontend builder (Codex) designs these from reference inspiration Yogi supplies, and Yogi approves them — the process is in A13
Which backend	The finished backend, as specified in Part D and delivered by the backend punch list. Nothing here describes a temporary state
Companion	HALO_Codex_Context.md — the whole-app briefing: the project, the repo as it stands, the backend's internals and status, every decision made so far, and the team workflow. Read it first, then this document
As of	September 27, 2026 — checked against main @ 45380c7 and the backend branch vmag211/backend @ 956f413

0. How to read this document

Every item is tagged so there's never a question of where a requirement came from or how firm it is:

Tag	Meaning	What to do with it
LOCKED	Functional or content requirement from the blueprint	Build exactly. Not up for redesign
BIND	The exact API field that feeds this element (full contract in Part D)	Bind to this field name, verbatim. Never compute the value yourself
FRONTEND RULE	A rule the frontend must follow because of how the API behaves — usually a trap that would silently show wrong information	Build it into the shared data layer, not per page
PROPOSED	Content or copy that isn't in the blueprint — usually extra richness from data the API already provides	Build it — every page should hold all the content it can — and list each PROPOSED item you built when you hand the page back, so Yogi can keep or cut it
DESIGN	A visual decision still to make	Decided in the page's design pass: Codex proposes from Yogi's reference inspiration and A11–A13; Yogi approves before the page is finished

THE ONE RULE THAT GOVERNS EVERYTHING BELOW — the server decides every severity level, every score, and every household-specific sentence. The frontend displays what it's given. It never compares a raw number against a threshold, never picks a severity word, never writes a personalized sentence of its own. Two screens computing the same threshold independently will eventually disagree about the same household (§6.1). Where this document says the frontend combines server values (e.g., "worst of four severity words"), it's choosing among words the server already produced, never deriving new ones.

BUILDING BEFORE THE BACKEND IS FINISHED — everything here targets the finished backend. If a page is built while some field isn't live yet, the absent-field rule (Part D, D0) covers it: a missing field shows as No data, an em dash, or a hidden element, never a crash. No page needs a special temporary version.

ON "EVERY PAGE SHOULD HAVE ALL THE CONTENT IT CAN" — for Today, HomeGuard, Journal, Act, Map, and the overlays, this document adds **PROPOSED** content wherever the API provides data the blueprint didn't use. **Onboarding is the deliberate exception:** the blueprint's own rule (§10.1) is that every extra question costs completions, and a household that abandons onboarding never sees any other page. Onboarding stays minimal on purpose; the richness goes into the pages after it.

Contents

- Part A** Cross-cutting systems — shell, navigation, data layer, severity, wording, formatting, copy, forms, accessibility, shared components, signature visuals, the locked design foundation (A11), priorities, performance, testing and limits (A12), the design process and the proposed base visual system (A13)
- Part B** Pages, in build order — B1 Onboarding · B2 Today · B3 HomeGuard · B4 Journal · B5 Act · B6 Settings · B7 Map and District panel
- Part C** Overlays — C1 Learn · C2 Assistant · C3 Alerts
- Part D** API reference — every endpoint and field the frontend binds to
- Part E** Definition of done, per page, and the build order
- Part F** Decision log — every product, design, and privacy decision made so far, and what's still open

Each page in Part B follows the same structure: purpose → address and entry → data calls in order → layout top to bottom → element-by-element table (content, binding, states) → page-level states → proposed content → visual decisions still to make.

Part A — Cross-Cutting Systems

Everything in this part applies to every page. Built once, in the shared layer, before any page — the blueprint's own first build step (§26.1) and the thing that decides whether dark mode and consistency are cheap or impossible later.

A1. Platform, shell, and navigation

Platform LOCKED

- A web application with a mobile-shaped layout, running in a mobile browser. Not native iOS or Android (§3).
- Primary viewport 375–430px. Every screen verified at 375px. Content column caps at 430px and centers above that; the Map is the one exception and widens to 600px (§53.2). Under 375px content compresses; nothing hides; no horizontal scroll ever.
- Portrait is the target. Landscape isn't blocked, but has no special layout (§53.4).
- Header respects the top safe-area inset, tab bar the bottom inset, sheets the bottom inset (§53.3).
- Stack: Next.js 16.2.10 App Router, React 19.2.4, Tailwind v4, `lucide-react` as the single icon family. The repo's `AGENTS.md` warns that this Next.js version has breaking changes from what most tooling assumes — whoever builds reads `node_modules/next/dist/docs/` before writing routing code.

Pages and overlays LOCKED

Page	Address	Tab label	Header title	Reached by
Onboarding	<code>/onboarding</code>	—	none (full screen)	Automatic on first visit
Today	<code>/today</code>	Today	Today	Tab 1, default landing
HomeGuard	<code>/home</code>	Home	Your Home	Tab 2
Map	<code>/map</code>	Map	Map	Tab 3
Journal	<code>/journal</code>	Journal	Journal	Tab 4
Act	<code>/act</code>	Act	Act	Tab 5
Settings	<code>/settings</code>	—	Settings (back arrow, not gear)	Gear icon in any header

Three overlays — Learn, Assistant, Alerts — are bottom sheets, never pages, never tabs. Learn opens only from a specific reading's "What does this mean?" link (no index, no browsing). Assistant opens from a floating button on all five tabbed pages. Alerts opens from the bell on Today's header only (§7.2).

Header bar LOCKED

Fixed top on every tabbed page. Page title left. Today: alerts bell (unread badge, capped at "9+", hidden at zero) and settings gear on the right. Every other tabbed page: gear only. Never hides on scroll (§7.3, §58.1). On Map, the header sits over the map (§58.1). `BIND` badge count = `GET /api/alerts` → `unread`.

Bottom tab bar LOCKED

Fixed bottom, five equal-width destinations, icon above label, active destination visibly distinct from inactive, always visible, never scrolls away (§7.4). Icons (§32.4): Today = sun over horizon, Home = house, Map = folded map, Journal = notebook, Act = outstretched hand. Tab changes are instant — no transition animation; content fades in as it arrives (§28.2). Each tab remembers its scroll position for the session; a newly visited tab starts at the top (§27.4).

Floating assistant button LOCKED

Lower right, above the tab bar, on all five tabbed pages. Hides while any sheet is open. Carries a chat-bubble icon (§32.4). The blueprint also has it fade out on scroll down and back on scroll up (§58.3) — **Tier 3: default is a fixed button** (§65.3). Opens the Assistant sheet carrying the current page as context.

First-visit routing LOCKED FRONTEND RULE

Blueprint (§7.5): on every load, check for a stored location. Have one → Today. Don't → onboarding. If the check itself fails → onboarding, never an error screen.

```
on app load:
  await ensureAnonSession()           // lib/auth.js – already built, AuthInitializer calls it
  GET /api/profile
    → onboarding_complete === true      → /today (or the requested deep link)
    → request failed, but local flag halo.onboarded is set → /today, retry the profile in the background
    → anything else                     → /onboarding
```

TWO RULES — (1) Route on `onboarding_complete`, not `onboarded`. `onboarded` turns true as soon as coordinates are saved — at onboarding's address step — so someone who closes the app on step 3 or 4 would be routed to Today with no water source, and a well household inside a utility boundary would be shown that utility's numbers. A household that didn't finish restarts at the address step (the address was never kept). When `GET /api/profile` succeeds with `onboarding_complete !== true`, clear `halo.onboarded` and every stored reading first — after a genuine 401 reset creates a new anonymous user, the old flag and cached readings would otherwise linger. (2) When onboarding finishes, write a local flag (`halo.onboarded = true`). If the profile request fails (network, server error) and the flag is set, go to Today and retry in the background — a finished household is never sent back through onboarding because of a failed request. The blueprint's "if the check fails, show onboarding" (§7.5) applies to households without the flag.

Addresses and state in the URL LOCKED

Address	Restores
---------	----------

/map?layer=radon	Map with radon layer active
/map?layer=water&contaminant=pfos	Map colored by PFOS
/map?system=NC0112010	Map with that system's sheet open
/journal?mode=trends	Journal in Trends mode
/home?risk=water	HomeGuard with the water card expanded (also radon , lead)
/today?learn=pfas	Today with that Learn sheet open

Not carried in the URL: scroll position, assistant conversation, unsaved journal input, which Settings accordion is open (\$27.2). /map?layer=water&quarter=2024Q3 applies only when the time slider exists (the map reports available quarters ; see B7). Any unrecognized address shows a short message and a button back to Today — never a raw error page, never a silent redirect (\$27.5).

The back-button contract LOCKED

- Opening any sheet (Learn, Assistant, Alerts, a map detail, any Act feature, Journal day detail, delete confirmation) pushes a history entry.
- Back with a sheet open closes the sheet and stays on the page beneath.
- Back with no sheet open goes to the previous tab, or exits.
- Closing a sheet any other way (drag down, tap backdrop, Escape, close control) also removes its history entry, so Back doesn't reopen it.
- Only one sheet at a time. Opening a second replaces the first.
- Expanding a card is not a history entry. Back never collapses a card.
- Each onboarding step pushes an entry; Back moves between steps without losing entered values.
- Opening a sheet locks page scroll beneath it; closing restores the exact prior position. Expanding a card scrolls just enough to reveal its new content (\$27.3-27.4).

Layering LOCKED

Back to front: page content → fixed header → fixed tab bar → floating assistant button → sheet backdrop → sheet → toast → offline banner. Nothing else creates a layer (\$53.1).

A2. The data layer

Authentication BIND

- `ensureAnonSession()` (lib/auth.js) — silent anonymous session, runs app-wide from the root layout. Throws with `.code` `'captcha_failed'` or `'signin_failed'`; the onboarding screen shows *"Verification could not be completed. Check your connection and try again."* or *"Could not start a session. Please reload and try again."*
- `authedFetch(url, init)` — attaches `Authorization: Bearer <token>`. Pass `init.headers` as a plain object (a `Headers` instance gets dropped by the spread).
- Every data route requires the bearer token. No route accepts identity from the body or query.
- 401 vs 503 — the rule that protects a household's data.** `401` means the session is definitively invalid (the account no longer exists): discard it and start a new anonymous session, silently (§31.4). `503` means identity couldn't be verified right now: show the generic error with retry and **never** reset the session. Resetting on a temporary failure would silently orphan the household's profile, history, and journal under an abandoned account.

One data hook, one state wrapper FRONTEND RULE

Every page gets data the same way — a single `useHaloData(endpoint, params)` hook returning `{ data, status, error, retrievedAt, fromCache, refresh }` and a single `<DataState>` wrapper that renders the right one of the four universal states. Pages never write their own loading/error logic. This is how "every screen implements all four states" (§24) actually holds across seven pages and two builders.

The four universal states LOCKED

State	When	Shows
Loading	Request in flight, nothing cached	Skeleton shaped exactly like the content it replaces — same dimensions, so nothing shifts when data arrives. Never a bare spinner (Map is the one exception, §46.9)
Error	Request failed	What failed + that it isn't a finding about their environment + Retry
Empty	Succeeded, nothing to show	Why there's nothing, and what would make content appear
Populated	Data present	The content

Error wording and empty wording are never the same text (§24). Every screen tolerates a missing water utility ID without breaking.

Error responses — what the frontend shows BIND

Response	Show
401 — any message	Nothing (silent re-auth, §31.4) — subject to the guard above
503 "Could not verify your session right now..."	Generic error + retry
429 (any route)	" <i>You've made a lot of requests. Please try again later.</i> " (Assistant has its own 429 text — see C2; pull to refresh has its own handling — next row)
429 on <code>/api/daily-score?fresh=1</code> — body <code>{ error, retry_after_seconds: 300 }</code>	Not an error state. The readings already on screen stay; the freshness line is unchanged; show a toast PROPOSED " <i>Your readings were just updated. Try again in a few minutes.</i> " The server allows one provider refresh per household per 5 minutes
400 on <code>/api/daily-score</code> — "No location yet..."	The household has no stored location: treat it exactly like <code>onboarding_complete: false</code> and route to onboarding (A1)
400 on <code>/api/home-guard</code> — "No county on file..."	Rare (the geocoder returned no county). HomeGuard shows its error state with a <i>Change address</i> action (B6); on the results reveal, the water and radon lines read <i>Couldn't check right now</i>
500, or network failure	" <i>Something went wrong on our end, not yours. Try again in a moment.</i> "
Timeout (client-side, 8s)	" <i>That took too long. The data source may be slow right now.</i> " Exceptions — the server doesn't enforce the blueprint's per-source timeouts yet (§62.2's 5s/8s/20s are targets; no provider call has a timeout in the code today), so a cold call can run long: <code>POST /api/onboard</code> , <code>GET /api/daily-score</code> and <code>GET /api/home-guard</code> use a 15s client timeout, and <code>POST /api/assistant</code> 25s (its timeout shows the Assistant's own "couldn't reach" message with Retry, C2). When <code>POST /api/onboard</code> times out on the client, re-check <code>GET /api/profile</code> before showing an error — the write may have succeeded; if a location is now stored, continue to step 3
Offline (<code>navigator.onLine</code> false)	Offline banner + last stored readings (below)

TWO BLUEPRINT STRINGS DELIBERATELY NOT USED — §31.4's "*[Source name] isn't responding right now. The rest of your readings are current.*" is dropped because §39.1 forbids naming services a household won't recognize — a missing reading shows its card's own no-data state instead (B2). §31.4's "*You appear to be offline. We're showing the last readings we have.*" is replaced by §58.4's banner text, "*You're offline — showing your last readings.*" Don't reintroduce either.

NEVER DISPLAY A SERVER'S error STRING DIRECTLY — most routes return raw database or JavaScript messages in 500s. Map every error to the copy above. The one exception is onboarding's specific address errors (B1), which are mapped by status code.

Stored readings, offline, and refresh LOCKED

- The last successful Today and HomeGuard responses are stored locally with the time they were received and shown immediately on open, while fresh data loads (§24, §36). Cached reading display target: under 200ms (§35.1).
- Fresh request fails and stored data exists → keep it on screen with "*Showing your last readings from [time]. We couldn't reach the data source.*"
- Offline → thin banner under the header: "*You're offline — showing your last readings.*" Pushes content down, never covers it (§58.4).
- Learn content cached after first read. Map dataset cached after first load (§36).
- Pull to refresh on Today and HomeGuard only. Today bypasses the server's one-hour cache (`/api/daily-score?fresh=1`); HomeGuard simply re-requests `GET /api/home-guard`, which has no server cache and no `fresh` parameter. Disabled while a request is in flight. Automatic refresh when the app regains focus after 15+ minutes away (§28.1). A refresh that returns cached data still updates the freshness line.
- Freshness lines use the server's `retrieved_at` (or `assembled_at`), never the time the browser received the response — a cached reading can be up to an hour old.
- Degradation order: fresh fails → stored data with its time → a single reading unavailable while others show → all fail: explanatory empty state with retry → server unreachable: stored data + offline banner (§39.2).

Journal writes offline TIER 3

The blueprint queues entries written offline and sends them on reconnect, with a small pending marker (§36). It's Tier 3 (§65.3): **the default is a clear failure instead** — keep everything typed and show PROPOSED "*Couldn't save that entry. It's still here — try again.*" Build the queue only if time allows.

A3. Severity and confidence

The six severity words LOCKED BIND

The backend emits exactly these words: `good`, `moderate`, `elevated`, `high`, `severe`, `no_data`. The frontend holds one mapping (word → label, icon, color token) — ideally importing the word list from `lib/severity.js` so it can't drift. Labels: Good, Moderate, Elevated, High, Severe, No data. Any unrecognized word renders as No data (§62.4).

- Every severity appears as **color + icon + word together**, never color alone (§6.1, §25.1). This is an accessibility requirement, not a style preference.
- No data is visually distinct from Good everywhere. Missing data is never shown as zero or green — the single most dangerous failure the blueprint names (§65.1).
- Pill labels capitalized; the same word lowercase inside a sentence (§29.3).

Where to read severity, per endpoint BIND

Screen element	Read	Never read
Today's four cards	<code>air.severity</code> , <code>uv.severity</code> , <code>pollen.severity</code> , <code>mold.severity</code>	<code>*.status</code> — older field with different thresholds (AQI 120 is <code>status "unhealthy"</code> but <code>severity "elevated"</code>)
Today trend lines, Journal grid	<code>/api/history</code> → <code>history[].air / uv / pollen / mold</code> (plain words)	—
HomeGuard cards & house diagram	<code>breakdown[].severity</code>	<code>breakdown[].status</code> — older field with its own thresholds
Both heroes (score ring, headline)	<code>score.severity</code>	—
HomeGuard action plan	<code>action_plan[].severity</code>	—
Lead card	<code>lead.level</code> (canonical)	—
Map	Water: <code>features[].overall_severity</code> , <code>features[].contaminants[X].severity</code> , <code>features[].hazard_index.severity</code> . Radon, facilities, air: <code>features[].severity</code>	—
District panel	<code>counties[].worst_water_severity</code> , <code>counties[].radon_severity</code>	—
Alerts	<code>alerts[].severity</code> (set for alerts that carry a level; else null)	—

Hero severity — what colors the score ring BIND

Both heroes read `score.severity` — the server decides the composite's level (§11.6). On HomeGuard it reflects water and radon only; **lead never colors the ring**, because it isn't scored. When the score is null (a private well, or nothing available), the ring shows its no-data state.

Confidence — a separate signal LOCKED BIND

value	Shown	Meaning
<code>full</code>	nothing	Everything evaluated, current
<code>limited</code>	quiet pill: <i>partial</i>	Some detections couldn't be evaluated
<code>stale</code>	quiet caution pill: <i>3+ yrs old</i>	Evaluated, but the sample is old
<code>none</code>	quiet pill: <i>not evaluated</i>	Nothing could be evaluated

HomeGuard's `breakdown[].confidence` carries this. A reading can be low severity and low confidence at once — never merge the two into one element (§6.2).

Provenance pills LOCKED

Three words: *measured*, *estimate*, *modeled* (§31.2). Air: `is_measured === true` → measured; `false` → modeled; `null` → no provenance pill (unknown isn't the same as modeled). Mold: always *estimate* (`is_proxy: true`). Radon: always *estimate* (county zone, not a home measurement). Lead: *estimate* when `lead.is_estimate` .

A4. Household-aware wording

LOCKED Seven yes/no facts per household, no ages, names, or medical records: `has_toddler`, `has_child`, `has_teen`, `has_adult`, `has_senior`, `has_pregnant`, `has_respiratory` (§8). Composition never changes a measurement, a severity, or a score — only the sentence under a reading and the order of actions (§8.2).

BIND The sentence is already written server-side: `air.sentence`, `uv.sentence`, `pollen.sentence`, `mold.sentence` (Today), `lead.sentence` (HomeGuard), `household_note` (Learn). Display them as-is. The frontend never composes a household sentence.

When several groups apply, the server uses the most specific one that has wording for that reading, in the blueprint's priority order: respiratory condition → toddler → pregnancy → senior → child → general (§8.5).

A5. Formatting rules LOCKED

One formatting module, used everywhere. The same quantity formatted two ways on two screens reads as unfinished (§29).

Quantity	Format	Example	Source field
Air quality index	Whole number	37	<code>air.aqi</code> (may be a decimal from the modeled source — round)
UV index	One decimal	5.2	<code>uv.index</code>
Pollen	Whole number with its category	Grass 3	<code>pollen.dominant</code> + <code>pollen[dominant]</code>
Concentration	One decimal, unit after a space	8.3 ppt	<code>value_ppt</code> (sent unrounded)
Ratio to limit	One decimal + ×	2.1× the limit	<code>value_ppt</code> / <code>limit_ppt</code> — arithmetic on server numbers, not a threshold
Composite score	Whole number, no unit, no %	78	<code>score.display_score</code> — never <code>score.score</code>
Radon zone	"Zone" + numeral	Zone 2	<code>radon.zone</code>
Population	Commas; rounded + "+" above 100,000	12,400 · 120,000+	<code>map</code> <code>population</code> , <code>district</code> <code>affected_population</code> , <code>population_over</code> . When <code>affected_population_complete</code> is false, say "at least"
Costs	Range, en dash, no decimals	\$40-90	<code>action plan</code> <code>cost</code> (already formatted); test plan <code>estimated_cost.low/high</code>
Percent	Whole number + %	78%	<code>journal</code> <code>elevated_rate</code>
Missing value	Em dash — never 0, never blank	—	<code>any</code> <code>null</code>

- Units spelled out the first time in a view, abbreviated after: "parts per trillion (ppt)" then "ppt."
- Measured values use tabular digits (Instrument Sans with `font-variant-numeric: tabular-nums`) so digits align — not a monospaced face. Monospace (IBM Plex Mono) is for codes only (A11).
- Never more precision than the source provides.

Dates LOCKED FRONTEND RULE

Situation	Format	Example
Under 1 hour	Relative minutes	updated 14 minutes ago
Under 24 hours	Relative hours	updated 6 hours ago
Yesterday	The word	yesterday
Last week	Weekday	Tuesday
This year	Month and day	March 14
Older	Month and year	October 2024
Lab sample dates	Always month + year, never relative	sampled October 2024
Data age warnings	Rounded years	this sample is about 3 years old

WHICH DATE FIELDS — water sample dates: bind the `*_iso` fields (`latest_sample_iso` , `earliest_sample_iso` , `date_iso`). The same objects also carry older `M/D/YYYY` strings (`latest_sample_date` , `date`) kept for compatibility — don't parse those; `new Date("7/1/2025")` is locale-dependent. Every other date is ISO.

ONE DATE CONVENTION — daily readings (`history[].date`) and journal entries (`entry_date`) are both the household's **local** calendar date. Join them by the same `YYYY-MM-DD` string. Compute "today" on the client in the household's local time, never from a UTC timestamp.

Text conventions LOCKED

- Sentence case everywhere — buttons and headings included. No title case, no all caps.
- PFOA, PFOS, PFHxS, PFNA, GenX keep their conventional capitals. The backend key `HFPO-DA` displays as "GenX (HFPO-DA)".
- Numbers under ten as words in prose, numerals in data. "County" always included.
- HALO always in capitals. Tagline where needed: "*Know what's around your home.*" Browser title pattern: "HALO — [page name]." Link preview description: "*See what's in the air, water, and ground around your home — and what to do about it.*" (§32).

Identity and icons LOCKED (§32)

- The logo (`/public/finallogo.png`) is used as: the Welcome mark (~120px wide); the browser tab icon at 32, 96 and 192px; the home-screen icon at 192 and 512px (a web app manifest); and the link-preview image at 1200×630 — the mark and the name over the primary color. Replace the default Next.js favicon.
- The browser `theme-color` is the Primary token — `#0e5e6f` in light mode, `#2cc4bd` in dark (two `<meta name="theme-color">` tags with `media` queries).
- One icon family (lucide-react), one line weight, no mixed styles. Concept icons: Today — sun over horizon · Home — house · Map — folded map · Journal — notebook · Act — outstretched hand · Air quality — wind lines · UV — sun · Pollen — flower · Mold — droplet · Water — tap · Radon — radiation trefoil · Lead — pipe · Alerts — bell · Settings — gear · Assistant — chat bubble. Where lucide has no exact match (e.g. a trefoil), pick the closest line icon at the same weight and keep it consistent everywhere.

A6. Voice and copy LOCKED

Plain, specific, steady. Never minimize a real problem; never frighten someone who can't act tonight (§30).

- Never "safe" or "unsafe" as a bare verdict. Never exclamation marks. Never fear words: toxic, poison, dangerous, alarming, shocking.
- Never apologize for data that doesn't exist. Never congratulate a good reading. Never "simply," "just," or "easy" for something that costs money or effort. Never imply a medical outcome.
- Second person ("your water"). HALO calls itself HALO, never "we" as an organization making promises.
- Reading level about sixth grade.
- Anything not listed in the content inventory (§31) and not derived from data doesn't appear in the interface. Where this document proposes new copy, it's tagged **PROPOSED** and needs approval.

BACKEND MESSAGES THAT BREAK THE VOICE RULES — USE BLUEPRINT COPY INSTEAD — some server `message` strings name services a household won't recognize (§39.1). Where the blueprint has specific copy for a state, use the blueprint's. Example: HomeGuard's `no_data_yet` — show the blueprint's *"Testing results for your utility haven't been published yet. This is normal — federal testing is still rolling out nationally."* instead. Each page section notes where the server's text is used and where the blueprint's replaces it.

DATA IN CAPITALS AND SERVER TEXT THAT BREAKS VOICE — utility names arrive as EPA and state records spell them, often in capitals (e.g. `"CONCORD, CITY OF"` in `lead.utility_inventory.utility`; `water.pws_name` likely the same). **PROPOSED** : display them in title case ("Concord, City of"), keeping acronyms such as WTP and state abbreviations in capitals. Don't build fixtures around the blueprint's "Concord — Hillgrove WTP", which is a treatment-plant name, not a utility record. A few server-written strings break A6 and are flagged for rewording by the content owner (Aadit) or Vibhav — e.g. a well test's `why` , "This is the one that can make your family sick this week, not in twenty years.", and alert titles like "Air quality dropped to Elevated today." (a capitalized severity mid-sentence). Display server text as sent; don't rewrite it in the frontend.

Universal copy **LOCKED**

Controls: Close, Back, Retry, Save, Cancel, Skip, Continue, Done, See more, Show less. Severity: Good, Moderate, Elevated, High, Severe, No data. Confidence: partial, 3+ yrs old, not evaluated. Provenance: measured, estimate, modeled (§31.1–31.2). Confirmations (§31.5): changing address — *"This will replace your current results. Continue?"*; delete all — *"This permanently deletes everything HALO knows about your household. This cannot be undone."* + typing a confirmation word; clear journal — *"This deletes all [n] of your entries permanently."*; delete one journal entry — immediate, with undo, no dialog.

A7. Forms and validation LOCKED

- Validate on blur, not on every keystroke. Error text directly beneath the field, field border also marked. An error clears the moment the input becomes valid.
- The submit button is never disabled for validation — let people submit and tell them what's wrong. (Different from disabling during an in-flight request, which is required.)
- Every field keeps its value after a failed submission. Every input has a visible label, not just a placeholder. Text inputs at 16px or larger (smaller makes mobile browsers zoom on focus). Address requests device autofill; build year opens the numeric keypad and accepts digits only (§34).

Field	Rule	Error text
Address	Required, 5+ characters	"Enter an address or ZIP code."
Address not found (server 404)	—	"We couldn't find that address. Try adding your city or ZIP code."
Build year	Optional; if present, four digits, 1700 to current year	"Enter a four-digit year, or leave this blank."
Build year in the future	Rejected	"That year hasn't happened yet."
Water source	Required, from the list	"Choose where your water comes from."
Household groups	Never required	—
Journal note	Optional, max 500 characters	Character counter appears past 400
Assistant question	Required, max 1000 characters	"Ask a question to get started."

BUILD YEAR — validate 1700 to the current year on the client (the rules above), and send it as a number, or `null` when blank. The server rejects values outside that range too, but the client check is what gives the person the right message.

A8. Accessibility, motion, text handling

Non-negotiable LOCKED

- Severity never by color alone. Every text/background pair meets standard contrast, verified separately in light and dark.
- Every interactive element at least 44×44px regardless of icon size.
- Everything reachable and operable by keyboard, in visual order, with a visible focus indicator. Escape closes any sheet. Enter/Space activates. Arrow keys move between segmented options and map layer chips.
- Sheets trap focus while open and return focus to whatever opened them.
- Icons and images have text alternatives; decorative ones are marked decorative. Every chart has a text equivalent stating the same finding.
- Body text never below the body size floor and respects the user's own text size setting.
- Hover states exist for pointer devices but never shift layout, and hover is never the only way to reveal something (§25.1, §28.6).

Motion — functional rules only LOCKED

Motion is limited to: sheet entry/exit, card expansion, content fade-in, press feedback, score ring sweep on first render, chart bar growth, onboarding step transitions, map color changes, toasts. Nothing longer than 400ms. Everything becomes instant under the system's reduced-motion setting, except opacity fades (kept very short so content doesn't pop). Exact durations and easing are DESIGN.

Touch and request feedback LOCKED

Every tappable element visibly responds on press within 100ms. Any button that triggers a request shows an in-place spinner and disables until it resolves (no double submits). Toggle chips flip instantly on tap, before the network call, and flip back visibly if the save fails (§28.5).

Text handling LOCKED

- Font stack declares a system fallback so text never waits on a font file.
- Card titles truncate at one line. Card summaries wrap to two lines then truncate. Utility names wrap to two lines in sheets, truncate at one in list rows. Journal notes truncate at one line in the list, full on tap. Assistant messages never truncate.
- **Numbers never truncate.** If a value doesn't fit, the container grows or the label shortens.
- Spanish runs 15–30% longer; no button width is fixed to its English label (§61).

Higher contrast (Settings) PROPOSED

The blueprint gives Settings a higher-contrast switch (§49.2) but defines no palette for it. Proposed effect: secondary text uses the body-text token, borders use Navy 700 (dark: a lighter border), and pills and focus rings get a 2px outline — implemented as a token override on the root, in both modes. Exact values DESIGN.

Dark mode LOCKED (colors DESIGN)

Follows the system setting automatically; no in-app toggle. Charts keep identical geometry with substituted colors. The health card's print output is always light. Build every color as a token from day one — the blueprint's own condition: if colors are hardcoded early, cut dark mode rather than retrofit it (§59, §68).

A9. The shared components — functional contract

Every page is assembled from these (§6.6, §44). This table is what each must *do* and which states it must support. What each *looks like* is **DESIGN**, decided on the first page that uses it and reused after.

Component	Must do	States / variants required	First page that needs it
Button	One primary per screen max	Primary, secondary, tertiary, destructive, icon-only × default, hover, pressed, loading (label → spinner, width held), disabled (only for genuinely unavailable actions, never validation)	Onboarding
Card	Container for content	Static, tappable, expandable, severity-marked (severity visible on the card edge or equivalent)	Today
Expandable card	Collapses to a summary row; expands in place; never navigates; not a history entry	Collapsed, expanded, loading, no-data	Today
Bottom sheet	Slides over current screen; dismiss by drag, backdrop tap, Escape, or close control; pushes a history entry; traps focus; locks scroll beneath	Heights: compact (≤50%, map details), standard (~70%, Learn/Alerts/Act), tall (~90%, Assistant). Internal scroll	Today (Learn)
Score ring	Shows a 0-100 composite, higher = better, colored by severity	Populated, no-data (no arc, em dash), partial (visibly incomplete without a caption), entrance sweep (skipped under reduced motion)	Today
Risk bar	Horizontal fill by risk, colored by severity	Populated, no-data, "shown, not scored" (no fill)	HomeGuard
Severity pill	Icon + word + color, always together	All six words	Today
Confidence pill	Quieter than severity; absent at full confidence	partial, 3+ yrs old, not evaluated	HomeGuard
Provenance pill	Same family as confidence	measured, modeled, estimate	Today
Callout	Inline note with an icon	Info, caution, notice tones	Today (incomplete-data notice)
Skeleton	Placeholder sized exactly like what replaces it	Animated, static (reduced motion)	Today
Empty state	Icon + heading + explanation + optional action	With / without action	Journal
Toast	Brief confirmation above the tab bar; one at a time	Plain, and with an Undo action. One duration for every toast , long enough to reach Undo (§65.3 — the blueprint's ~4s/~8s split is Tier 3)	Journal
Tab bar	Five destinations	Active, inactive, pressed	Shell
Header	Title + right controls	Today variant (bell + gear), standard (gear), Settings (back arrow), Map (over map)	Shell
Segmented control	2-3 options, switches content, no navigation	Journal mode, journal severity, letter recipient, owner/renter	Journal
Toggle chip	On/off label; flips instantly	Selected, unselected, disabled	Onboarding (household)
Switch	Settings rows	On, off, disabled	Settings
Accordion row	Settings sections; one open at a time; current value summary when closed	Collapsed, expanded	Settings
Text / numeric input, dropdown, checkbox, slider	Per A7	Default, focused, error, disabled	Onboarding
Offline banner	Under header, pushes content down	Shown / hidden	Shell
Swipe action	Reveals delete/dismiss; action is immediate with an undo toast	Journal entry delete, alert dismiss	Journal

THE BLUEPRINT'S OWN SIMPLIFICATIONS ARE THE DEFAULT (§65.4) — where the table above lists more variants than a page needs, build the simpler set unless a page genuinely requires more: **buttons** — three variants (primary, secondary, tertiary); destructive is secondary in the Elevated red, icon-only is tertiary with no visible label (it keeps an accessible name). **Sheet heights** — two (standard ~70%, tall ~90%); map details use standard. **Cards** — mark severity with the pill, not a colored card edge as well ("both is redundant emphasis"). **Skeletons** — two shapes, a card block and a text block, composed as needed. **Motion** — three durations: 150ms fades, 250ms movement, 400ms score ring.

A10. The signature visuals — what's locked vs. what's open

The blueprint's exact pixel geometry for these (§56) is visual and void. What they must *do* is not — each carries meaning, which is why they exist (§6.0).

Visual	Locked function LOCKED	Open DESIGN
Score ring (Today, HomeGuard hero)	0-100, higher is better, same direction on every screen. Fill = score/100. No-data = no arc + em dash. Partial = visibly incomplete, distinguishable without a caption. No "out of 100," no percent sign.	Size, stroke, how "partial" looks, entrance

		motion
Contribution bar (under each hero)	One stacked bar; segments sized by each factor's share of the total risk, colored by that factor's severity, each labeled (tiny segments labeled only in a legend). Tap a segment → scroll to and expand its card.	Thickness, labels, legend treatment
Threshold bar chart (HomeGuard water, map sheets, health card)	One row per contaminant. Scale is always 0 to 2× that contaminant's federal limit, so the limit sits at the same position on every row — this is what makes different contaminants comparable. Values past 2× fill fully with a visible "continues" end and the true multiple in the label; never rescale the axis. Limit marked on every scored row. Sort worst-first by ratio. Unscored substances below a divider, de-emphasized, no limit marker, labeled "shown, not scored." Tap a row → Learn for that contaminant.	Row/bar dimensions, marker style, entrance motion
House diagram (HomeGuard)	Three zones — tap (water), foundation (radon), pipe (lead) — each shown in its severity, grey when no data. Each zone is a tap target that scrolls to and expands its card. Schematic, not an illustration. If it turns into an illustration project, replace with three labeled rows carrying the same colors — the page loses character, not meaning.	Everything about how it's drawn
Calendar heat grid (Journal)	One cell per day, seven columns. Cell = that day's worst severity. No-data day visibly different from Good. Future days not drawn. Logged-symptom days marked. Today marked. Tap a day → that day's detail sheet.	Cell size, colors, markers
Seven-day trend (Today expanded cards)	Daily values, today emphasized, severity bands behind the line so a rise visibly enters a worse band. Missing days break the line — never interpolated.	Size, curve, labels
Quarterly trend (HomeGuard water)	Worst contaminant's readings across sampling dates, federal limit as a reference line. Only drawn when there are 2+ dated readings.	Everything visual
Comparison bar pair (district panel, map banner)	Two bars on the same scale.	Everything visual

Every chart has a plain-language caption stating its takeaway, units on any axis, missing data as a visible gap (never zero, never interpolated), a screen-reader text equivalent, and colors from the severity scale only — no chart gets its own palette (§6.8). All drawn as inline SVG, no charting library (§35.2).

A11. The design foundation — locked

Decided once, used by every page. The blueprint's own color and type values (\$6.3, \$6.4, \$59) are replaced by these.

Severity colors LOCKED

Word	Label	Light mode	Dark mode	Style	Icon
good	Good	#1e7a3c green	#4cc27a	Solid fill	Circle with check
moderate	Moderate	#b85c00 orange	#f0913a	Solid fill	Circle with i
elevated	Elevated	#c0262d red	#f0605f	Solid fill	Triangle with exclamation
high	High	#8e2a6b red-purple	#d668ad	Solid fill	Octagon with exclamation
severe	Severe	#4f1440 deep purple	#a870f0	Solid fill	Filled octagon
no_data	No data	#111111 black	#e6e6e6 near-white	Outline only — never filled	Circle with question mark

No data is drawn as an outline so it can never be mistaken for Severe (black and deep purple are nearly the same brightness). Pill text on a filled level is white in light mode and navy (#081725) in dark mode. Every color above passes 4.5 : 1 for text in its mode (checked by calculation). Icons have clearly different shapes because severity is always color + icon + word.

Brand and surface colors LOCKED

Token	Light	Dark	Use
Navy 900	#0b1f33	#081725 (page)	Main text in light mode; the base of dark mode
Navy 700	#133b5c	#1d3a50 (border)	Headers, deep accents
Primary — deep teal	#0e5e6f	#2cc4bd	Primary buttons, active tab, links, focus ring
Teal 500	#138a8a	—	Secondary accents — never text (fails contrast)
Turquoise	#1fb5b0	—	Highlights and fills only — never text
Light aqua	#a8e6e2	#a8e6e2 on #1d4f5c	Selected chips, soft tints — the light accent, used sparingly
Page background	#eef8f7 aqua mist	#081725	Behind cards
Card surface	#ffffff	#0f2436	Cards and sheets
Border	#d3e4e6	#1d3a50	Dividers, card outlines
Body text	#0b1f33	#e8f3f5	All primary text
Secondary text	#4a6275	#9bb4c2	Timestamps, sources, captions

No saturated green anywhere in the brand, so green always means Good. Light mode is the default; dark mode follows the system setting.

Typefaces LOCKED

Role	Face	Notes
Headings, hero score numbers, page titles	Fraunces (variable)	A soft serif — the app's personality. Has no evenly spaced digits, so an animated score needs a fixed-width container so it doesn't wobble
Everything else: body, labels, buttons, cards, data rows	Instrument Sans	Readings use its tabular digits (<code>font-variant-numeric: tabular-nums</code>) so columns align — verified the font includes them
Codes only	IBM Plex Mono	Water system IDs (<code>NC0190010</code>), certifications (<code>NSF/ANSI 53</code>). Not for readings

All three are free Google Fonts, loaded through Next.js's built-in font loading with a system fallback so text never waits (\$61). All include the accented characters Spanish needs. A proposed starting type scale is in A13; it's finalized in Onboarding's design pass and then used everywhere.

REPLACED, NOT TO BE USED — everything visual in the blueprint (\$6.3–6.5, \$56–60 pixel values), Vibhav's glassmorphic HomeGuard branch and its docs/HALO-design-language.md (Playfair Display + Poppins, dark gradients, glass nav, the plexus ban), the current onboarding screen's look (aurora and plexus canvases, frosted glass card, "Proxima Soft"/"DM Sans"), and the Geist fonts and Arial body in today's app/layout.tsx and globals.css . The foundation above replaces all of them.

A12. Priorities, performance, testing, and limits

The blueprint specifies everything; it also says completeness is not priority (§65). This section is how to spend effort. Read it before building.

Tier 1 — load-bearing, never cut LOCKED (§65.1)

Item	Why
The four universal states on every screen	A screen that silently shows nothing is indistinguishable from a broken one
Error copy distinct from empty copy	"We failed" and "the data doesn't exist yet" need completely different reactions
Severity as color + icon + word	A risk app must not fail a colorblind reader
Missing data never shown as zero or green	The single most dangerous failure — absence presented as safety
Every estimate labeled	The app's credibility rests on not overstating what it knows
The back-button contract (A1)	Without it overlays feel broken on every device
Number and date formatting (A5)	The same value formatted two ways reads as unfinished
The shared component set, built first	Otherwise every page looks like a different app
Bind to the frozen contract (Part D)	A shape change after screens exist blocks everyone
Dark mode	Ignoring the system preference reads as unfinished. Cheap only because every color is a token from day one
Health card print styling (B5)	Its stated purpose is to be handed to a doctor or landlord
Assistant's structural refusal to diagnose (server-side)	A prompt can be argued around live; a code path can't
Threshold bar's fixed 0–2× limit scale (A10)	What makes different contaminants comparable at a glance
"Felt fine today" in Journal (B4)	Without good days the pattern analysis is biased
Journal's minimum data before reporting	Four data points is the difference between credible and misleading
Stored readings shown when a source fails (A2)	An environmental app that goes blank on a weak signal fails when it's wanted

Tier 2 — build in this order once Tier 1 is solid LOCKED (§65.2)

1. The map's contaminant selector — the most interactive element, the one most likely to hold a judge's attention
2. Household-aware wording (display the server's sentences — A4)
3. The threshold bar chart
4. The Journal calendar heat grid
5. The HomeGuard house diagram
6. The results reveal after onboarding
7. The contribution bar
8. Seven-day trend lines
9. Pull to refresh
10. Spanish (only once a person has reviewed the translation — §68)

Tier 3 — cut first, in this order, without hesitation (§65.3)

Assistant button hide/show on scroll (leave it fixed) · map auto-panning above a sheet · the home pin's pulse · score number counting up · chart bars staggered · skeleton shimmer (a static block is fine) · header shadow on scroll · onboarding slide transitions (a cross-fade or nothing is fine) · air-monitor glow · hiding zoom buttons on touch · toast duration varying by content (one duration) · the map time slider (only if the data supports it — §68) · season recap notifications · landscape handling · offline journal queueing (a clear "couldn't save, try again" is acceptable).

Page cut order if time runs short (§26.3): Assistant sheet → Settings → Act → district panel. Cutting Settings means household groups can only be set in onboarding, and renting mode can't be set at all — onboarding has no renting question (B1), so every household gets owner wording. If Settings is cut, keep HomeGuard's PROPOSED "Showing actions for renters · Change" control above the action plan (B3) and have it save `renter_mode` through the one `PUT /api/household` save function (all seven groups). Two finished pages beat five half-finished ones; an unfinished page reads as broken rather than scoped.

Performance targets LOCKED (§35)

Measure	Target	Maximum
First meaningful content	under 1.5s	3s
Interactive, responding to taps	under 2s	4s
Tab change	under 100ms	300ms
Sheet open	under 250ms	—
Cached reading display	under 200ms	500ms
Fresh reading retrieval	under 2s	5s, then show stored readings with the notice (A2)
Map first render	under 2s	4s
Map layer switch	under 300ms	1s
Total page weight, first load	under 500KB	1MB

Measured on a mid-range phone over a mobile connection. How: charts are inline SVG (no charting library); the map library loads only when the Map tab is first opened; map data is one pre-built file per layer; stored readings render immediately while fresh data loads; images sized for display in a

modern format; fonts load with a fallback shown immediately; skeletons occupy exactly the space of what replaces them (§35.2–35.3).

Security and privacy in the client LOCKED (§37)

- No analytics or tracking SDK of any kind — and the privacy statement says so (§37.4).
- No secret in browser code. The only browser-visible values are the Supabase URL and anon key, the optional Turnstile site key, the VAPID public key, and — only if the map uses Mapbox GL JS — a public `pk.` map token restricted to the app's URLs. The server's `MAPBOX_TOKEN` (geocoding) is never exposed.
- The journal note and the assistant question are the two free-text inputs: render them as text, never as HTML.
- Browser storage holds only the session (managed by the auth library), stored Today and HomeGuard readings, cached Learn content and map data, the onboarded flag, device-local accessibility settings, small interface memory (dismissed seasonal prompts, the last-used journal group), and any queued journal entries — nothing a person wouldn't already see on screen.
- Never send a household or profile id as identity; the token is the identity.

Testing — what every page must survive LOCKED (§40)

Test addresses (real, not synthetic)	Must produce
A Concord address	A utility match with PFAS above the federal limit — the primary demo case
A Union County address	A different utility with different readings
A third NC-08 address	A third utility, ideally with no published results (the no-data state)
A rural address	No utility match — the well prompt
An address entered as a well	The testing plan, never a score
An address outside North Carolina	Daily readings work; home readings degrade gracefully (radon <code>out_of_state</code> ; water says HALO's water data covers NC — B3)
A Buncombe County address (Zone 1) and an Iredell County address (Zone 2) — in NC but outside NC-08	The radon-elevated paths: a radon action item, the <code>radon</code> volunteer cause, Act's radon help section, Journal's winter radon prompt, the <code>radon_season</code> alert, and a well plan's radon-in-water test

EVERY NC-08 COUNTY IS RADON ZONE 3 — in the stored EPA data, every county in and around NC-08 (Cabarrus, Union, Stanly, Anson, Montgomery, Richmond, Moore, Rowan, Mecklenburg, Scotland, Robeson, Hoke) is Zone 3. So for every NC-08 demo address the radon card reads Good · Zone 3, radon never enters the action plan, and the radon prompt, alert, and help section never appear. Test those paths with the Zone 1/2 addresses above; use Zone 3 in NC-08 fixtures. "Zone 2" examples in this document are illustrative.

Cases: no household groups (general wording everywhere) · every group selected · respiratory only (wins the priority) · build years before 1950, 1950–1987, 1988 or later, and blank · renting vs owning (every action changes) · Journal with 0, 4 (below threshold), and 20 entries including illness-flagged days · every map layer and contaminant, including one with no readings anywhere · the Assistant against ten adversarial questions including direct requests for diagnosis · every empty and error state triggered on purpose · both languages on every screen.

Technical: every screen at 375px on a real phone (not a resized desktop window) · both mobile engines (Safari/WebKit and Chrome) · keyboard-only through every screen · a screen reader on Today and HomeGuard at minimum · one data source deliberately blocked · fully offline, then back online · Back with every sheet type open · deep links pasted into a fresh browser.

Before any demo (§40.4): run `/api/health` (it needs a session token — call it with `authedFetch` from the running app's console) · load the demo address and confirm results within target · confirm the map data was rebuilt recently · have a fallback if a government service is down.

What HALO deliberately cannot do — each is stated in the interface LOCKED (§69)

Limitation	Where the interface says it
It can't tell you what's in your tap water — only what the utility measured at its own sampling points	HomeGuard water card (B3)
It can't measure radon in your home — county zones predict averages	HomeGuard radon card (B3)
It can't detect mold — it estimates favorable conditions from humidity and rain	Today mold card, "estimate" pill (B2)
It can't test private wells — no agency does	Well testing plan (B3), results reveal (B1)
It can't tell you whether your plumbing contains lead — age indicates an era, not a material	HomeGuard lead card (B3)
It can't diagnose anything	Assistant disclaimer and refusal (C2)
It can't prove an exposure caused a symptom	Journal findings disclaimer (B4)
It can't promise its data is current	Every reading carries its date (A5)
It can't cover the whole country equally — water and radon detail is strongest in North Carolina	Results reveal note for non-NC addresses (B1)

What the judges actually see (§65.5)

Whether it works at all, on a real device, without errors · the map, especially the moment a contaminant selection recolors the state · whether numbers are explained in language a person understands · whether it admits what it doesn't know · whether the presenter can explain why it was built this way. They won't notice easing curves or a two-pixel shadow. "If a trade-off ever arises between looking finished and being honest, choose honest."

A13. The design process, and the proposed base visual system

Who designs, and how LOCKED

Codex does the wireframes, the visual design, and the code for every page. Yogi supplies reference inspiration — screenshots of real apps whose feel he wants — and approves each page's look. There is no separate mockup step: the "wireframe" is the real page, coded at 375px with fixture data, which then gets wired to the API. Per page:

- Read the function first.** The page's section in Part B, plus every shared part it uses (A1–A13, C, D). The element table is the checklist: every element, binding, state, and copy string.
- Study the references Yogi supplies.** Borrow what they do well — hierarchy, spacing rhythm, density, how a card or chart is treated. Don't borrow their colors or fonts (A11 governs), and don't borrow their emptiness if this page holds more data than theirs.
- Build the page against fixture data covering every state** — populated, loading, empty, error, partial/no-data, offline — in light and dark, at 375px.
- Check it against the element table, line by line, independent of the look.** A visual change never drops an element, a state, or a string to look cleaner. If a reference's layout can't hold a required element, the layout changes, not the requirement.
- Show Yogi screenshots of every state.** Iterate on visuals only. List every PROPOSED item built.
- Wire to the live API (Part D)** and run the definition of done (E1).

The quality bar — high-end, not overboard

The blueprint's own bar (\$6.0, \$65.5): clean, deliberate, finished — not a funded consumer product. What reads as high-end is discipline: one type scale, one spacing scale, one color system applied predictably, aligned edges and even margins, text that never overflows, nothing shifting as content loads, correct contrast in both modes, and precise details (consistent icon stroke weight, optical alignment of icons with text, tabular digits in data). Personality comes from the signature elements, one per page, each carrying meaning (\$6.7): Today's time-of-day gradient behind the hero, HomeGuard's house diagram, the Map's full-bleed canvas, Journal's heat grid, Act's editorial cards, Learn's typographic sheet. Buttons, cards, and type stay identical everywhere. "If a visual decision takes more than an hour and doesn't change what a person understands, it isn't worth making."

What makes an interface look generated — avoid all of it

- Gradient washes, glassmorphism and background blur, decorative blobs or particle/plexus canvases (the only gradients are Today's time-of-day hero and Onboarding's Welcome band)
- Drop shadows on every card; colored left-border cards; everything a rounded pill
- Emoji as icons; mixed icon families (lucide-react only, one stroke weight); icons drawn for decoration
- Colors outside the A11 tokens; saturated green anywhere except the Good severity
- Center-aligned body text; title case or all caps; exclamation marks; lorem ipsum or invented numbers
- A fake phone frame or status bar — the app is the real page at the real device width (the current onboarding screen's fixed 390×844 shell must go)
- Uneven spacing, near-miss alignments, animation on everything

Proposed base visual system PROPOSED

A starting point from the Onboarding planning session, so the first page doesn't start from nothing. Codex refines it in Onboarding's design pass with Yogi's approval; once approved it's used on every page unchanged.

Type role	Face	Size / line height	Use
Hero score	Fraunces 600	56 / 60	The composite number on Today and HomeGuard — fixed-width container (Fraunces has no tabular digits)
Display	Fraunces 600	30–32 / 38	Onboarding Welcome sentence
Step / page heading	Fraunces 600	26 / 32	Onboarding step questions
Header title	Fraunces 600	22 / 28	Page titles in the header bar
Section heading	Fraunces 600	18 / 24	Section dividers within a page
Card title	Instrument Sans 600	16 / 22	Card headers
Body large	Instrument Sans 400	17 / 24	Supporting sentences, privacy line
Body	Instrument Sans 400	15 / 22	All explanatory text — the floor ; nothing readable is smaller (\$25.1)
Small	Instrument Sans 500	13 / 18	Timestamps, sources, captions, field labels, error text — never explanations
Pill label	Instrument Sans 600	12 / 16	Severity, confidence, provenance pills only
Button	Instrument Sans 600	16	All button labels

Element	Proposed treatment
Spacing	4px grid: 4, 8, 12, 16, 24, 32, 48. Page gutter 20px
Radius	Cards and inputs 14px · buttons 12px · chips and pills fully rounded · sheets 20px on the top corners
Surfaces	Cards: card color + 1px border, no shadow. The primary button is the one element with a soft shadow (0 2px 8px rgba(11,31,51,.12))
Primary	Primary teal fill, white text, 52px tall, full width in forms. Pressed: 1px down. Loading: width held, label replaced by an in-place spinner

button	
Secondary	Transparent, 1.5px primary border, primary text. Destructive = secondary in the Elevated red
Tertiary	Primary text only, no border or fill (e.g. "Skip this"). Icon-only = tertiary, 44×44 target, accessible name
Disabled	40% opacity, no shadow — only for genuinely unavailable actions, never validation (A7)
Text input / dropdown	52px tall, 1px border, 14px radius, card fill; a visible label above the field (Small, secondary text). Focus: 2px primary border plus the keyboard focus ring. Error: Elevated-red border and red error text with an icon beneath — the field fill never turns red
Toggle chip	Unselected: card fill, 1px border, body text. Selected: light-aqua tint, 1.5px primary border, and a small check icon, so the state isn't carried by color alone; <code>aria-pressed</code>
Onboarding progress	Four 3px segments with 4px gaps, filled primary / unfilled border color, no numbers shown; screen-reader text "Step 2 of 4"
Severity pill	Icon + word, fully rounded, filled in the level's color with white text (light) or <code>#081725</code> text (dark); No data outline only
Motion	150ms fades, 250ms movement (sheets; onboarding steps as a short slide or a cross-fade), 400ms score ring. Reduced motion: instant, except very short opacity fades

Part B — Pages

B1. Onboarding

Purpose LOCKED

Collect the absolute minimum needed to produce a real result, and nothing more. No account creation, no email, no password, no permission prompt — the anonymous account already exists before this screen renders (§10.1). A household that finishes onboarding lands on real, personalized results; one that abandons it sees nothing. That's why this is the one page that stays deliberately short.

Address and entry

`/onboarding`, reached automatically by the first-visit routing in A1 (profile says `onboarding_complete` isn't true, or the profile request failed and the local `halo.onboarded` flag isn't set), directly for testing, or from Settings → Your home → Change address (starts at step 2). Full screen: no header, no tab bar, no assistant button (§51.1).

What exists today vs. what this page must be

CURRENT CODE — `app/OnboardingScreen.tsx` (699 lines) is a single-screen form: address + build year + water source, styled with animated canvas backgrounds and glass cards. **Keep:** `ensureAnonSession` bootstrap and its error messages, the GPS handling (sends coordinates as `"lng,lat"` — its old fallback to a hardcoded Austin address has already been removed, and that must not come back), the `"lng,lat"` encoding for device location, build-year validation (range changes to 1700). **Replace:** every visual, the single-screen structure (becomes four steps), the generic error handling (becomes four distinct messages), the dead end at submit (a `TODO` and `console.log` — it never routes anywhere). **Add:** the Welcome step, the Household step (doesn't exist at all), the privacy line, the lead-solder note, the results reveal.

Structure LOCKED

Four full-screen steps. A slim four-part progress indicator at the top (not shown on Welcome). A back arrow from step 2 onward returns to the previous step with every entered value intact. Each step is a history entry. Each step: the question, any supporting line, the input, then the primary button at the bottom (§10.2, §51).

Data calls, in order BIND

When	Call	Sends	Why here
Page load	<code>ensureAnonSession()</code>	—	Already runs app-wide; onboarding surfaces its failure
Step 2 "Continue"	<code>POST /api/onboard</code>	<code>{ address }</code>	So address problems surface on the address step, where the blueprint puts them (§10.4)
Step 3 "Continue" (not on Skip)	<code>PUT /api/household</code>	all seven booleans, every time	Saves composition. Skip sends nothing, leaving <code>household_set: false</code>
Step 4 "See my results"	<code>PATCH /api/profile</code>	<code>{ water_source, home_year }</code>	Saves the home details without re-sending the address — the address was used once at step 2 and never kept
Results reveal	In parallel: <code>GET /api/daily-score</code> and <code>GET /api/home-guard</code>	No parameters — both use what onboarding just stored. The onboard response's <code>state</code> decides the outside-NC note	Fills the four reveal lines and warms the cache for Today and HomeGuard

TRAP — `PUT /api/household` is a full replace: any of the seven not sent become `false`. Always send all seven, as real booleans.

Step 1 — Welcome LOCKED

Element	Content	Notes
Mark	HALO logo	Exists at <code>/public/finallogo.png</code> (§32.1)
Sentence	"HALO shows you what's in the air, water, and ground around your home — and what to do about it. No account needed."	§10.3
Primary button	<i>Get started</i>	→ step 2
Supporting line	"Takes about a minute."	Below the button
Session failure	"Verification could not be completed. Check your connection and try again." or "Could not start a session. Please reload and try again."	From <code>ensureAnonSession</code> <code>error.code</code> ; Get started disabled until a session exists

Step 2 — Address LOCKED

Element	Content / behavior	Binding / notes
Heading	"Where do you live?"	
Text field	Label visible; placeholder "Street address or ZIP code"; device address autofill; 16px+ text	Validation on blur: 5+ characters else "Enter an address or ZIP code."
Secondary	<i>Use my location</i>	<code>navigator.geolocation</code> → field shows "Using your current location"; sends <code>"lng,lat"</code>

button		as <code>address</code> . Unsupported: " <i>This device can't share its location. Please type your address.</i> " Denied/failed: " <i>Couldn't get your location. Please type your address instead.</i> " Typing in the field discards the coordinates
Privacy line	" <i>We use your address once to find your local data, then throw it away. We keep only your approximate coordinates, your county, and which water utility serves you — never the street address itself.</i> "	Directly under the field, visible before they type (§10.4)
Primary button	<i>Continue</i> ; while processing: loading state + supporting text " <i>Finding your local data.</i> "	Calls <code>POST /api/onboard { address }</code>

Outcomes, by response **BIND**

Response	Show	Next
200	—	Step 3. Keep <code>service_area_status</code> , <code>county</code> for later
404	" <i>We couldn't find that address. Try adding your city or ZIP code.</i> " Field keeps its contents	Stay
429 (either message)	" <i>You've made a lot of requests. Please try again later.</i> " No blame, no detail	Stay
500 / network	" <i>Something went wrong on our end, not yours. Try again in a moment.</i> " + Retry	Stay
401 / 503	Handled by the auth layer (A2)	—

OUTSIDE NORTH CAROLINA — a non-NC address proceeds normally (§10.4). The onboard response carries `state` ; when it isn't `NC` , the results reveal adds a note that daily air, UV, and pollen readings work everywhere while water and radon detail is strongest in North Carolina. **PROPOSED** wording: "*Daily air, UV, and pollen readings work anywhere. Water and radon detail is strongest in North Carolina.*"

NOTE — `service_area_status: "outside_known_area"` is a normal, common result in NC-08 (no utility boundary contains the point) — not an error. `lookup_failed` means the boundary lookup itself failed; onboarding still continues. Either way, step 4's water-source question matters more for these households.

Step 3 — Household **LOCKED**

Element	Content / behavior	Binding
Heading	" <i>Who lives here?</i> "	
Supporting text	" <i>This lets HALO tell you what today's readings mean for the people in your home. We never ask for ages, birthdays, or names — just which groups apply.</i> "	
Seven toggle chips, wrapping, this order, all start unselected	Toddler · Child · Teen · Adult · Senior (65+) · Someone pregnant · Someone with asthma or a breathing condition	<code>has_toddler</code> , <code>has_child</code> , <code>has_teen</code> , <code>has_adult</code> , <code>has_senior</code> , <code>has_pregnant</code> , <code>has_respiratory</code>
Primary	<i>Continue</i>	<code>PUT /api/household</code> with all seven. On failure: keep selections, show the generic error, allow Continue again
Skip	<i>Skip this</i> — same row as Continue, equal visual weight so skipping never feels like failing	No call. General wording everywhere; never prompted again; editable in Settings

Step 4 — Your home **LOCKED**

Element	Content / behavior	Binding
Heading	" <i>A little about your home.</i> "	
Build year	Label " <i>Year built</i> ", optional, numeric keypad, digits only. Under it: " <i>Homes built before 1988 may have lead solder in their plumbing. If you're not sure, leave it blank — we'll tell you we don't know rather than guess.</i> "	<code>home_year</code> : number or <code>null</code> . Validation per A7 (1700–current year)
Water source	Label " <i>Where does your water come from?</i> "; options: City or town water · Private well · Spring · Other · Not sure. Required	<code>water_source</code> . The server normalizes these exact labels: City or town water → <code>utility</code> , Private well → <code>well</code> , Spring → <code>spring</code> , Other and Not sure → <code>other</code>
Primary	<i>See my results</i>	<code>PATCH /api/profile { water_source, home_year }</code> , set the local onboarded flag, go to the results reveal. On failure: keep both values, show the generic error, allow retry

This one choice is the largest branch in the application: well or spring sends all of HomeGuard to a testing plan instead of measurements (§10.6).

The results reveal **LOCKED**

A transitional screen after step 4, before Today (§33, §51.5): the logo, then four lines appearing in sequence, each starting with a small working indicator that turns into a check as its result resolves, with the result on the right. Then a primary button *See my results* → `/today` . Under reduced motion all four appear at once. Purpose: makes the work visible, introduces the four things HALO tracks, and turns a redirect into proof the app did something real.

The four line labels and the example results come from §33. Result strings not in the blueprint are marked **PROPOSED** .

Line	Result shown	Binding

<i>Finding your water utility...</i>	Utility name, e.g. <i>Concord — Hillgrove WTP</i> · or PROPOSED <i>No utility matched your address</i> · <i>Private well — no utility / Spring — no utility</i>	<code>water.pws_name</code> ; <code>water.status = no_pwsid_available / private_well</code> (+ <code>water.source_type</code>). For <code>no_data_yet</code> and <code>lookup_failed</code> there's no name in the response — show PROPOSED <i>Found your water system ([pwsid])</i> from the onboard response's <code>pwsid</code>
<i>Checking federal testing results...</i>	<i>PFOS found above the limit</i> (§33) · <i>No published water results yet for your utility.</i> (§33) · PROPOSED : <i>Nothing found above federal limits</i> · <i>Wells aren't tested by any agency</i> · <i>Found compounds with no federal limit</i> · <i>Couldn't check right now</i>	Measured: the worst entry in <code>water.scored_contaminants</code> with <code>exceeds_limit: true</code> (the server decides — the frontend never compares a value to a limit); none → "Nothing found above federal limits." Other states by <code>water.status : no_data_yet</code> , <code>private_well</code> , <code>no_pwsid_available</code> ("No utility to check"), <code>detected_unregulated</code> , <code>lookup_failed</code>
<i>Looking up radon for [County]...</i>	Zone 2 (§33's example — every NC-08 county is actually Zone 3, A12) · or PROPOSED <i>Not available for this county</i> · for out-of-state addresses PROPOSED <i>Covers North Carolina only</i>	<code>radon.zone</code> ; <code>radon.error</code> present → not available; <code>radon.out_of_state</code> → the out-of-state line
<i>Getting today's air quality...</i>	Severity label, e.g. <i>Good</i>	<code>air.severity</code> → label

If a lookup produced nothing, the line says so plainly rather than disappearing (§33). For an address outside North Carolina (`state` isn't `NC`), a `no_data_yet` water result reads **PROPOSED** *HALO's water testing data covers North Carolina utilities* instead of the "not published yet" line, which would be false there. If a call fails outright, its line reads *Couldn't check right now* and the button still appears — Today and HomeGuard have their own error states.

VOICE CHECK — "Nothing found above federal limits" is a statement about a measurement, which the rules allow. Never "Your water is safe" (§30.3).

ROUTING — the local `halo.onboarded` flag is set only after step 4 succeeds. The server's `onboarding_complete` turns true at the same moment (coordinates and water source both stored); `onboarded` turns true earlier, at step 2, which is why routing never uses it (A1).

Proposed content **PROPOSED**

None beyond the outside-NC note. This page is deliberately minimal — see the note in Part 0.

Visual decisions — made and still open

This is the first page built, so its design pass finalizes the base visual system (A13) that every later page reuses.

Decision	Status
Palette and typefaces	LOCKED — A11. Light mode default, dark follows the system
Overall tone	DECIDED — "something in between" quiet and immersive: the page background (aqua mist) everywhere, with a brand moment on Welcome only. Steps 2–4 and the results reveal stay light and calm so fields, errors, and results read clearly
Welcome's brand moment	DECIDED (F15) — the navy-to-teal band behind the logo, on Welcome only. PROPOSED values: a band about 220px tall across the top of Welcome, a gradient from Navy 900 <code>#0b1f33</code> to Primary teal <code>#0e5e6f</code> at about 165°, with the logo centered in it. Below it the page background resumes for the sentence, the button, and "Takes about a minute." This band and Today's time-of-day hero are the only gradients in the app
The logo	<code>/public/finallogo.png</code> — 906×818 transparent PNG: a circular badge with a house/shield outline holding a family of four, ringed by four small icons (cloud for air, drop for water, leaf/spore for pollen and mold, sun for UV), in teal, cyan, and amber. It needs a treatment that holds up on the navy band — e.g. sitting on a white or light-aqua disc, or a one-color white variant — DESIGN . The results reveal uses the standard full-color logo, small
Base components	PROPOSED — buttons, inputs, dropdown, toggle chips, progress segments, type scale, spacing, radius, and motion as in A13. Finalized here
Step transition	PROPOSED — a 250ms horizontal slide (Back reverses it) or a cross-fade; instant under reduced motion. Tier 3 — a cross-fade or nothing is acceptable (§65.3)
Results reveal	PROPOSED — each of the four lines fades up (short rise + opacity) in sequence over about two seconds (§33); each line's small working indicator becomes a check when its result arrives, or a neutral dash when the result is "not available". Indicators drawn in the app's icon stroke weight. Reduced motion: all four lines appear at once with their final icons
Still open DESIGN	Exact band height and gradient angle, the logo treatment on navy, the Welcome sentence's size and line breaks, chip wrapping and spacing on Household, the dropdown's open state, the error-state styling in context, and the reveal's exact timing

WHAT MUST GO FROM THE CURRENT SCREEN — the fixed 390×844 phone shell (the page must fill the real device width, A1), the aurora and plexus canvases, the frosted glass card, the undefined "Proxima Soft"/"DM Sans" fonts, the old tagline "Know what surrounds your family." (Welcome uses the locked sentence; the app's tagline, where one is needed, is "Know what's around your home.", §32), the non-link "Terms of Service" footer, and the single-screen form.

B2. Today

Purpose LOCKED

Answer, in under two seconds of looking: *is it a good day to be outside, and is there anything I should do differently today?* This is the page a household opens habitually — legible at a glance, and it never requires scrolling to deliver its headline (§11.1).

Address and entry

`/today` — default landing for onboarded households, Tab 1. `/today?learn=<topic>` opens that Learn sheet over it.

Data calls BIND

Call	Params	Feeds
<code>GET /api/profile</code> (from app boot)	—	<code>profile.county</code> , <code>household</code>
<code>GET /api/daily-score</code>	none — the server uses the household's stored location. <code>?fresh=1</code> on pull to refresh	Hero, four cards
<code>GET /api/history?days=8</code>	—	Trend lines; yesterday's score for the proposed change line
<code>GET /api/alerts</code>	—	Bell badge (<code>unread</code>)

DON'T SEND COORDINATES — call `/api/daily-score` without `lat` / `lng` so it uses the household's stored location. Readings for any other coordinates (a GPS fix, a rounded value) are fetched but never saved to the household's history or cache.

Layout, top to bottom LOCKED

- Header: *Today* · alerts bell with unread badge · gear
- Hero block (roughly the top third, no scroll needed): score ring → headline → notices → location/freshness line → contribution bar, on the time-of-day background
- Four reading cards, fixed order: Air quality, UV index, Pollen, Mold risk. Order never changes — habitual users navigate by position (§11.4)
- Floating assistant button

Hero LOCKED

Element	Content	Binding	States
Score ring	The composite, 0-100, higher is better	<code>score.display_score</code> (never <code>score.score</code> , which is a two-decimal float; never <code>score.risk</code> , which runs the other way)	Partial when <code>score.is_partial</code> . No-data when <code>display_score</code> is null — and see the guard below
Ring color / headline level	—	<code>score.severity</code>	No data when null
Headline	By level: Good → " <i>Good day to be outside</i> " · Moderate → " <i>A mostly good day, with one thing to watch</i> " · Elevated → " <i>Worth taking some care outside today</i> " · High or Severe → " <i>A rough day out there — see what's driving it</i> " · No data → " <i>We couldn't get readings for your area today</i> "	From the level above	
Incomplete notice	" <i>Based on 3 of 4 readings — pollen data isn't available right now.</i> " The score still shows; it's never presented as complete when it isn't	Count = <code>score.included_inputs.length</code> ; names from <code>score.missing_inputs</code> (air → "air quality", uv → "UV", pollen → "pollen", mold → "mold")	Only when <code>is_partial</code>
Estimate notice	" <i>Includes an estimated value, not only measurements.</i> "	<code>score.includes_proxy === true</code>	Nearly always true (mold is always an estimate)
Location line	" <i>Union County · updated 14 minutes ago</i> "	<code>profile.county</code> + <code>retrieved_at</code> as relative time (A5)	Manual refresh that returns cached data still updates this line
Contribution bar	What's driving today's score, without opening a card	Segment size = each factor's share of the total of <code>score.inputs.air/uv/pollen/mold</code> (these are risks, higher = worse); segment color = that factor's <code>.severity</code> . Factors with <code>null</code> input are absent	Tap segment → scroll to + expand that card
Time-of-day background	Changes with the device clock — before dawn, midday, dusk, night. Atmospheric only; carries no information; text contrast verified in every state	Device clock	Concept LOCKED (§6.7, §11.3 — Today's signature element); the color stops are DESIGN , drawn from the A11 tokens, with text contrast verified in every state

WHEN EVERY SOURCE FAILS — the server returns `display_score: null` with every input in `missing_inputs` and `reason` set. The hero shows its No data state and the headline "We couldn't get readings for your area today" — never a number built from nothing.

The four reading cards LOCKED

Collapsed (every card): severity icon, card name, measured value (tabular digits, A11) with unit, severity pill, any provenance/confidence pills, the one-sentence household-aware explanation beneath, and a chevron. **Expanded:** the detail below, and a *"What does this mean?"* link that opens Learn for that topic.

Card	Collapsed value	Pills	Sentence	Expanded contents
Air quality	<code>air.aqi</code> , whole number, unit "AQI". Null → em dash	<code>air.severity</code> ; <i>measured</i> if <code>is_measured</code> true, <i>modeled</i> if false, none if null	<code>air.sentence</code>	7-day trend (history <code>air</code> words + <code>values.aqi</code>) · provenance line: true → <i>"Measured at a monitoring station near you"</i> ; false → <i>"Modeled estimate — the nearest monitoring station is more than 25 miles away."</i> ; null → PROPOSED <i>"The source of this reading wasn't recorded."</i> · static note: the index reflects the single worst pollutant, not an average · the dominant pollutant with a one-line explanation of what it is (<code>air.dominant_pollutant</code> : "PM2.5" → fine particles, "PM10" → coarse particles, "O3" → ozone — the frontend holds the plain-language line for each; hidden when null) · Learn: <code>topic=air</code> with its context (C1)
UV index	<code>uv.index</code> , one decimal. Null → em dash	<code>uv.severity</code>	<code>uv.sentence</code>	7-day trend · expected peak hours today, e.g. <i>"Strongest 11am-3pm, peaking at 7.4"</i> (<code>uv.peak_window.start</code> , <code>.end</code> , <code>.max</code> ; hidden when null) · static note: UV passes through cloud cover · Learn: <code>topic=uv</code> with its context (C1)
Pollen	The worst category named: <code>pollen.dominant</code> + its value, e.g. <i>Grass 3</i> . All null → em dash	<code>pollen.severity</code> (severity of the dominant category)	<code>pollen.sentence</code>	All three categories: <code>pollen.tree</code> , <code>.grass</code> , <code>.weed</code> each with its own value and severity pill (<code>pollen.categories.tree</code> / <code>grass</code> / <code>weed</code> → <code>value</code> , <code>severity</code>) · 7-day trend of the dominant category · seasonal note by month (static, NC: tree late winter-spring, grass into summer, ragweed in fall) · Learn: <code>topic=pollen</code> with its context (C1)
Mold risk	No numeric value exists — show <i>Estimate</i> in the value position (DESIGN)	<code>mold.severity</code> + <i>estimate</i> always	<code>mold.sentence</code>	Always, without exception: <i>"This is an estimate based on humidity and rainfall forecasts, not a sensor in your home."</i> · the humidity and chance of rain the estimate is built from (<code>mold.basis.humidity_pct</code> , <code>.precip_pct</code>) · static: what would change the estimate · Learn: <code>topic=mold</code> with its context (C1)

NEVER BIND status — each reading object also carries a `status` field (`unhealthy` , `very_high` , `low` , `none` ...) from older code with different thresholds. It disagrees with `severity` for many real values. Only `severity` exists as far as the UI is concerned.

MOLD NEVER READS HIGHER THAN ELEVATED — the server caps it on purpose because it's an estimate. When the weather forecast couldn't be retrieved, mold is `no_data` with `risk: null` — not a quiet "low."

Page-level states

State	Shows
First load, nothing stored	Skeleton: ring-shaped placeholder, headline bar, location bar, four card-shaped blocks, same sizes as the real content (§52)
Stored readings exist	Show them immediately with their original time; fetch fresh in the background
Fresh fetch fails, stored exists	Keep stored + <i>"Showing your last readings from [time]. We couldn't reach the data source."</i>
Fails, nothing stored	Empty state: <i>"We couldn't get readings for your area right now. This is usually temporary — pull down to try again."</i> + Retry
One reading missing	That card shows No data and its server sentence (e.g. <i>"We couldn't get a UV reading for your area right now."</i>); the other three render normally; the hero shows the incomplete notice
Trend with under 2 days of history	<i>"Your trend will appear here after a few days of readings."</i>
Offline	Banner + stored readings

Interactions **LOCKED**

- Pull down to refresh (disabled mid-request) → `/api/daily-score?fresh=1` , bypassing the server's one-hour cache. The server allows one fresh fetch per household every 5 minutes; a repeat inside that window returns 429 { `retry_after_seconds: 300` } — keep the current readings on screen and show the short toast from A2, never the error state.
- A 400 *"No location yet..."* means there's no stored location — route to onboarding (A1).
- Auto-refresh on returning to the app after 15+ minutes away.
- Tap a card → expand/collapse in place (not a history entry). Tap a contribution-bar segment → scroll to and expand that card.
- Bell → Alerts sheet. The badge shows the unread count and clears as items are read (**PROPOSED** — §54 says opening the sheet "clears the badge"; that would need a mark-all-read on open, which erases which alerts are new. Yogi's call). Gear → Settings. Assistant button → Assistant with `page=today` .

Proposed content **PROPOSED**

Addition	From	Why
Change since yesterday under the ring, e.g. <i>"6 lower than yesterday"</i>	<code>history</code> — yesterday's <code>score</code> vs today's <code>display_score</code> ; hidden if yesterday is missing	Answers "is today worse than usual" without opening anything
A "worded for your household" line, e.g. <i>"Readings explained for: a toddler, someone with asthma · Edit"</i>	<code>household</code> from profile; hidden if <code>household_set</code> is false	Makes the personalization visible — a judge sees immediately that the sentences are tailored, and the household can fix it if wrong
A week summary inside each expanded card, e.g. <i>"2 of the last 7 days were Elevated or worse"</i>	<code>history</code> severity words, counted	Turns the trend line into a sentence (every chart needs a takeaway caption anyway, §6.8)

Provenance explained once in the air card: measured vs modeled, and that roughly half the district has no monitor within 25 miles	Static, from \$41.2	This disclosure is unusual for a consumer app and worth making visible (\$11.4)
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Visual decisions to make DESIGN

First use of: the six severity colors and icons (Phase 0 — locked once for the whole app), score ring (including the partial and no-data looks), contribution bar, severity/provenance pills, expandable card anatomy, seven-day trend line, callouts, skeletons, the header with bell badge, the tab bar, the floating assistant button. Also: the time-of-day background's color stops, and the hero's composition.

B3. HomeGuard

Purpose LOCKED

Answer *what is true about this specific building, and what should we do about it?* These facts change over years, not hours, so the page is read carefully rather than glanced at — it can carry more depth and more text than Today (§12.1). It should read as a building inspection, not a weather report.

Address and entry

/home , Tab 2. /home?risk=water|radon|lead opens with that card expanded.

Data calls BIND

Call	Params	Feeds
GET /api/profile	—	pwsid (shown with the utility name), home_year , renter_mode
GET /api/home-guard	none — the server uses the stored county, utility, water source, and build year	Everything on the page

DON'T PASS THE HOME'S DETAILS — call /api/home-guard with no parameters so it always reflects what's stored. (The endpoint also accepts county / pwsid / water_source / home_year overrides; no page uses them.) Store the last response locally so it shows instantly on the next open (A2).

Response shape at a glance BIND

```
{ success, radon, water, score, lead, action_plan, breakdown }
  water → five shapes, discriminate on water.status (table below)
  score → three shapes: private well (no score) · neither input available · normal
  lead → object for public water; null for well/spring
  action_plan → [] for well/spring
  breakdown → [water box, radon box, lead box (public only), one box per excluded substance (lithium)]
```

Layout, top to bottom LOCKED

- Header: *Your Home* · gear
- Hero: score ring → headline naming the dominant contributor → basis line → contribution bar (water vs radon)
- House diagram
- Three risk cards, fixed order: Drinking water, Radon, Lead
- Section heading *Your Action Plan* → numbered items (or the testing plan's summary for wells)
- Floating assistant button

Hero LOCKED

Element	Content	Binding / rule
Score ring	Home score 0-100, higher is better	score.display_score . Ring color: score.severity — water and radon only, never lead (A3). Null in two cases: private well/spring (score.reason : "No score can be calculated for a private well or spring...") and no inputs at all. Show the no-data ring with the reason text beneath — a missing score here is an expected state, not an error. Partial when score.is_partial
Floor	—	The score never goes below 6 (score.floored). A 6 means "as bad as it gets," not "6% good." No special copy needed; don't treat it as near-zero
Cap notice	When the water result couldn't be fully evaluated, the score is capped (70 or 85) and says why	Show score.reason when non-null (the partial / unscoreable explanation strings)
Headline	Names the dominant contributor, e.g. "Water is driving most of this" / "Radon is driving most of this"; or "Nothing here needs urgent attention."	Driver = larger of score.water_risk and score.radon_risk (comparing two server numbers). "Nothing urgent" when action_plan is empty and the source isn't a well
Basis line	"Based on EPA testing of your water utility and county radon data."	Well/spring variant PROPOSED : "Based on county radon data. No agency has tested your well water."
Contribution bar	Water vs radon share of the home risk	Sizes from score.water_risk , score.radon_risk ; colors from each box's severity . Lead is never in this bar — it isn't scored. Tap → scroll to that card

House diagram LOCKED

Three zones — tap (water), foundation (radon), pipe (lead) — each in its severity, grey with no data. Each zone taps to scroll to and expand its card. Purpose: shows three unrelated readings as facts about one building, and visually separates this page from Today (§12.2b). Zone severity: water and radon from their breakdown[].severity ; lead from lead.level . Well households: lead zone grey with the lead card explaining why (below).

Card 1 — Drinking water LOCKED

Collapsed: name, risk bar (fill from the water box `risk` , 0–100), severity pill, confidence pill from box `confidence` , and one line naming the driving fact.

Expanded — the deepest single view in the app (\$12.3). Contents depend entirely on `water.status` :

<code>water.status</code>	What it means	What the card shows
anything else (e.g. <code>measured</code>)	Real federal testing results exist	The full measured view — next table
<code>detected_unregulated</code>	Compounds were detected but none has a federal limit or benchmark	The measured view, but the threshold chart has no scored rows. Lead with <code>water.message</code> (ends "Detected is not the same as safe.") and the detected names from <code>water.detected_unregulated</code> . Score is capped — show <code>score.reason</code>
<code>private_well</code>	No agency has ever tested this water	The testing plan instead of any score (below)
<code>no_data_yet</code>	Valid utility, no published results	Neutral, not a warning: " <i>Testing results for your utility haven't been published yet. This is normal — federal testing is still rolling out nationally.</i> " (blueprint copy — not the server's message, which names the dataset). Show the utility's ID (<code>profile.pwsid</code>) so the household can check it's theirs — there's no name in this response. Outside North Carolina (<code>profile.state</code> isn't <code>NC</code>): HALO's water data covers NC utilities only, so this copy would be false — show PROPOSED " <i>HALO's water testing data covers North Carolina utilities.</i> "
<code>lookup_failed</code>	Something broke on our side	" <i>This is a problem on our end, not a finding about your water.</i> " + Retry. Radon and lead still render. Never the same wording as <code>no_data_yet</code> — one means wait, the other means try again (\$12.4)
<code>no_pwsid_available</code>	Address outside every mapped utility boundary	" <i>We couldn't match your address to a water utility. Are you on a private well?</i> " + choices (PROPOSED labels) <i>Private well</i> · <i>Spring</i> · <i>No, city or town water</i> . Well/Spring → save it with <code>PATCH /api/profile { water_source }</code> , then reload HomeGuard, which now shows the testing plan. "No" → no change; the card keeps the prompt's explanation

Measured view, in order BIND

Element	Content	Binding
Utility	Name and ID, so the household can check it's the right one	<code>water.pws_name</code> + <code>profile.pwsid</code>
Threshold bar chart	One row per scored contaminant, worst first; unscored below a divider (A10)	Scored rows: <code>water.scored_contaminants[]</code> → <code>contaminant</code> , <code>value_ppt</code> , <code>limit_ppt</code> ; <code>ratio</code> = <code>value/limit</code> ; sort by <code>ratio</code> . Below the divider: <code>water.excluded_from_score[]</code> (lithium, with its value) and <code>water.detected_unregulated_latest[]</code> → <code>contaminant</code> , <code>value_ppt</code> , <code>date_iso</code>
Detail rows	Per scored contaminant: name, value with units, the limit, the ratio, date sampled. Worst first	<code>value_ppt</code> , <code>limit_ppt</code> , <code>ratio</code> , <code>date_iso</code> (→ "sampled July 2025")
Limit type	Per row: enforceable federal limit vs. a health benchmark that isn't enforceable	<code>is_enforceable</code> , <code>basis</code> , <code>source</code> . Rows scored against a benchmark: label "health benchmark — not an enforceable limit"
Rescission marker	Per row whose limit is proposed for repeal	<code>proposed_for_rescission</code>
Quarterly trend	Worst contaminant across sampling dates with the limit as a reference line — "is it getting better?"	<code>water.contaminants[<worst name>]</code> = every reading as <code>{ date_iso, value_ppt }</code> . Draw only with 2+ dated readings; plot by each reading's <code>date_iso</code>
Unregulated group	Detected, no enforceable limit — shown, with the statement that these are not enforceable limits	<code>water.detected_unregulated</code> ; <code>water.message</code> when present
Guidance-scored group	Scored against a health benchmark rather than a federal limit	<code>water.guidance_scored[]</code> → <code>contaminant</code> , <code>limit_ppt</code> , <code>basis</code> , <code>source</code> ; <code>water.includes_guidance</code>
Excluded from score	Lithium: measured, shown, not scored, with the reason written out	<code>water.excluded_from_score[].reason</code> (verbatim), <code>value_ppt</code> , <code>date_iso</code> , <code>health_reference_source</code> ; the lithium box adds <code>health_reference_level_ug_l</code> and <code>detail[].value_ug_l</code>

Regulatory notice	Callout when the score rests on a limit proposed for rescission	<code>water.regulatory_notice</code> (present only when it applies). The server's wording names the specific compound ("The federal limit for PFHxS is currently the subject of a proposed EPA rescission..."); the blueprint's example is general ("Three of the five federal PFAS limits have been proposed for rescission. This score uses one of them.", §12.3). Recommendation: use the server's — it's more specific and stays correct as the rules change
Sample age	Caution callout when the latest sample is roughly 3+ years old: <i>"This sample is about 3 years old."</i>	Water box <code>confidence === "stale"</code> (server decides), <code>data_age_days</code> , <code>latest_sample_iso</code>
Learn link	<i>"What does this mean?"</i> → PFAS	<code>topic=pfas&contaminant=&value=&limit=</code> from the worst scored row

Private well / spring — the testing plan LOCKED BIND

No score, ever — inventing a reading for untested water would be dishonest (§12.5).

Element	Binding
Framing	<code>water.message</code> — the well version ("Private wells are not covered by the Safe Drinking Water Act...") or the stronger spring version, chosen by <code>water.source_type</code>
Why these tests	<code>water.test_plan.reasons[]</code> — already tailored to the county's radon zone, the build year, and well vs spring
Tests, in priority order	<code>test_plan.tests[]</code> → <code>rank</code> , <code>name</code> , <code>priority</code> (critical / recommended / optional), <code>cadence</code> , <code>why</code> , <code>where</code> , <code>estimated_cost.low - .high</code>
Totals	<code>test_plan.start_here_cost</code> (all critical tests) and <code>test_plan.full_panel_cost</code> , as ranges

NOTE — test costs are estimates (`estimated_cost.is_estimate: true`, same on both totals) — label them as such.

Card 2 — Radon LOCKED

Element	Content	Binding
Collapsed	Name, risk bar, severity pill, <i>estimate</i> pill, one line e.g. <i>"Zone 2 · a county average, not your home"</i>	Radon box <code>risk</code> ; severity per A3; <code>radon.zone</code>
County and zone	<i>Union County · Zone 3</i> (every NC-08 county is Zone 3 — A12)	<code>radon.county</code> , <code>radon.zone</code>
What a zone is and isn't	<i>"Zones predict the average indoor level across a whole county. They are not a measurement of your home. Homes in low-zone counties still test high, and neighboring houses can differ."</i>	Static, §12.3
Action threshold, testing, cost	The federal action level (4 pCi/L), what a test involves, rough cost	Static; cost matches the radon action item (<i>\$15–40 for a test kit</i>)
Seasonal note	October–February only: winter readings are the most accurate (sealed, heated homes draw in more soil gas)	Device month
County not found	PROPOSED <i>"Radon zone data isn't available for your county."</i>	<code>radon.error</code> present, no <code>out_of_state</code>
Outside North Carolina	PROPOSED <i>"HALO's radon zone data covers North Carolina addresses. A home test is the only way to know your level anywhere."</i> Card in its no-data state	<code>radon.out_of_state === true</code> (also carries <code>radon.state</code> , e.g. <code>"SC"</code>). Never show an NC zone for an out-of-state county with a matching name
Learn	<i>"What does this mean?"</i>	<code>topic=radon&county=&zone=</code>

Card 3 — Lead LOCKED

Informational only. It has a severity for display and ordering and can produce an action item, but it never contributes to any score — settled, final (§12.3). The card header carries a permanent *shown, not scored* label, matching lithium.

Element	Content	Binding
Collapsed	Name, <i>shown, not scored</i> , severity pill, <i>estimate</i> pill when estimated, one line	<code>lead.level</code> , <code>lead.is_estimate</code> , <code>lead.contributes_to_score</code> (always false). Risk bar: no fill
Household sentence	e.g. <i>"...Young children absorb lead far more readily than adults, so this is worth testing sooner rather than later."</i>	<code>lead.sentence</code>
Basis statement	Every age-based result says it's an estimate from the home's age, not a measurement of the plumbing	<code>lead.text</code> (includes that sentence)
By build year	Before 1950 → High · 1950–1987 → Elevated · 1988 or later → Good · unknown → No data	<code>lead.basis</code> : <code>built_before_1950</code> , <code>built_1950_1987</code> , <code>built_1988_or_later</code> , <code>unknown_year</code>
Unknown year	<i>"We don't know when this home was built, so we can't estimate lead risk. Testing is the only way to know."</i> + link to add the build year in Settings. For an unknown year, <code>lead.sentence</code> , <code>lead.text</code> and this copy are the same sentence — show it once . In general, render <code>lead.text</code> only when it differs from <code>lead.sentence</code>	<code>lead.basis === "unknown_year"</code> . Saved via <code>PATCH /api/profile { home_year }</code> (B6)
Private well	PROPOSED <i>"For a private well, lead is part of your testing plan above."</i>	<code>lead === null</code>

Learn	"What does this mean?"	topic=lead&home_year=
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DECIDED (backend, recorded) — THE SEVERE LEAD LEVEL IS KNOWINGLY UNREACHABLE — the Severe level (`basis: "utility_lead_service_line"`) would need to know that this specific address has a lead service line. No public address-level inventory exists for NC-08 utilities, so the backend never produces it. Lead is always age-based (High, Elevated, Good, or No data), with the estimate pill whenever `lead.is_estimate` is true (A3). The mapping still handles `severe` in case that changes.

Element	Content	Binding
Your utility's service lines	PROPOSED Context under the lead estimate, only when the utility appears in the state's published inventory: " <i>[Utility] reported [needs_replacement] service lines that need replacing (lead or galvanized) and [unknown] of unknown material. This is for the whole utility, not your address.</i> " + source line " <i>NC DEQ lead service line inventory, July 2025</i> ". It never changes the lead level	<code>lead.utility_inventory</code> : { <code>utility</code> , <code>reported</code> , <code>lead</code> , <code>galvanized_requiring_replacement</code> , <code>non_lead</code> , <code>unknown</code> , <code>needs_replacement</code> , <code>address_level</code> : <code>false</code> , <code>source</code> : { <code>name</code> , <code>url</code> , <code>retrieved</code> } } or null (hide the row). Format counts with commas (A5)

Your Action Plan **LOCKED** **BIND**

Numbered, worst problem first, only items at Elevated or above. Each item expands to its full guidance (\$12.6, \$54).

Element	Binding
Order and number	<code>action_plan[].rank</code> (the server already moves water and lead up when a toddler or pregnancy is present)
Severity	<code>action_plan[].severity</code>
Title	<code>title</code> , e.g. "Filter your drinking water for PFOS"
Reason naming the finding	<code>reason</code> , e.g. "PFOS measured at 8.3 ppt — about 2.1× the federal limit of 4 ppt." Show as sent — the value in it is already rounded to one decimal
Cost range	<code>cost</code> (already formatted, e.g. "\$30-90")
Certification to look for	<code>certification</code> (null for radon) — shown inline, so it can be read standing in a store
What to do	<code>action</code> — already written for renter or owner by the server
Renter / owner tag	From <code>profile.renter_mode</code> — the text is already tailored; the tag says for whom
Empty	" <i>Nothing here needs action right now. We'll tell you if that changes.</i> " — but never for a well : a well's action plan is always empty because the testing plan replaces it. Show a pointer to the testing plan instead

VERIFY BEFORE LAUNCH — every certification number (NSF/ANSI 53, P473, NSF/ANSI 58, NSF/ANSI 42) against the certifying body's own documentation (\$12.6, \$66). A wrong filtration claim is materially harmful to a household acting on it.

Page-level states

State	Shows
Loading, nothing stored	Skeleton: ring, headline, house-diagram block, three card blocks
Request fails (500)	Stored response if any + notice; else " <i>Something went wrong on our end, not yours. Try again in a moment.</i> " + Retry
Radon unavailable	<code>radon.error</code> present (county not matched, or <code>out_of_state</code>) — the radon card shows its no-data state; water and lead render normally

Proposed content **PROPOSED**

Addition	From
A "what this is based on" line in the water card: " <i>Based on 6 readings from March 2023 to July 2025</i> "	<code>water.reading_count</code> , <code>earliest_sample_iso</code> , <code>latest_sample_iso</code>
Detected count: " <i>4 compounds detected · 3 scored · 1 with no federal limit</i> "	<code>water.detected_count</code> , <code>scored_count</code> , <code>detected_unregulated.length</code>
Renter/owner line above the action plan: " <i>Showing actions for renters · Change</i> "	<code>profile.renter_mode</code> → Settings
A small "what HomeGuard can't tell you" footer: what's at your tap depends on your own plumbing; zones aren't home measurements; build year isn't a pipe inspection	Static, from \$69 — the blueprint calls stating limits plainly an asset in judging

Visual decisions to make **DESIGN**

House diagram (the page's signature), risk bar, confidence pill, threshold bar chart, quarterly trend, testing-plan list, action-plan item anatomy, first use of the bottom sheet if Learn wasn't built on Today, and how this page reads as "inspection" rather than "weather."

B4. Journal

Purpose LOCKED

Connect environmental readings to how the people in the household actually feel. The only feature where the household puts information in and gets something specific back — what separates a tool from a display (§14.1).

THREE BINDING RULES FROM THE RETENTION RESEARCH (§14.2) — (1) The app must be fully useful to someone who never logs anything — Journal is optional depth, never a prerequisite. (2) No streaks, badges, points, or celebration animations. (3) The first insight must arrive immediately, not after two weeks.

Address and entry

`/journal` (Log mode), Tab 4. `/journal?mode=trends` opens Trends.

Data calls BIND

Call	Params	Feeds
GET <code>/api/profile</code>	—	household → group selector
GET <code>/api/journal</code>	default = last 90 days	Entries: <code>entries[]</code> → <code>id</code> , <code>entry_date</code> , <code>band</code> , <code>symptoms[]</code> , <code>severity</code> , <code>note</code> , <code>retrospective</code> , <code>possibly_illness</code>
GET <code>/api/history?days=90</code>	—	Heat grid, day detail, each entry's day condition, retrospective result, line chart
POST <code>/api/journal/retrospective</code>	<code>{ dates }</code>	The first-visit result: <code>{ ready, factor, statement }</code>
GET <code>/api/journal/findings</code>	—	Trends findings or not-ready progress
POST <code>/api/journal</code>	see the form table	Save / update / "Felt fine today" / retrospective days

Shape LOCKED

Header *Journal*. Directly beneath, a two-option segmented control: *Log · Trends*. Switching changes the content below without navigating (§14.3).

Log mode — first visit (no entries at all) LOCKED

Element	Content / behavior	Binding
Prompt	"Think back over the last few weeks. Which days were rough — headaches, congestion, trouble sleeping? Tap them."	Shown when <code>entries</code> is empty
Month calendar	Weekday initials header; past days selectable, future days dimmed and not selectable; month arrows, back no more than three months	—
Count	"3 days selected."	—
Primary	See what was in the air — active at three or more days; below that, helper text explains why	Saves each selected day with <code>POST /api/journal</code> and <code>retrospective: true</code> — see the rule below
Result	The calendar collapses and a callout appears, e.g. "On the 3 days you flagged, grass pollen averaged High. The month's average was Low. Days you didn't flag averaged Low." Then a tertiary Start logging daily	<code>POST /api/journal/retrospective { dates }</code> → <code>statement</code> — see below

HOW THE RETROSPECTIVE WORKS — after saving the flagged days, call `POST /api/journal/retrospective` with `{ dates: [...] }`. The server compares the flagged days' readings with the rest of that month and returns `{ ready, factor, statement }` — the frontend shows `statement` as-is and claims nothing more (§14.4). `ready: false` means too few of those days have readings; show its `statement` the same way. The flagged days are saved with `retrospective: true`, `band: "household"`, and `symptoms: ["unspecified"]` — the findings analysis counts a day as a symptom day only if `symptoms` is non-empty, and the person didn't say which symptoms.

WHY THIS WORKS ON A FIRST VISIT — the server records a day's readings for every household every day, and fills in recent weeks at onboarding where a historical source exists, so the flagged dates usually have readings before the person ever logs anything (§14.4). Dates without readings (older than the backfill, or a day a source had nothing) are left out of the comparison.

Log mode — ongoing LOCKED

Opens straight to today's form once any entry exists. The whole interaction must fit in under ten seconds (§14.5).

Element	Content / behavior	Binding (POST body)
Date header	"Today, March 14" with left/right arrows to log an earlier day (no future days)	<code>entry_date</code> — YYYY-MM-DD, the household's local calendar date (joined to history by the same string — see the date note in A5)
Who	Toggle chips naming each group present; hidden entirely for a single-group or skipped household; last-used group preselected	<code>band</code> — the group key (e.g. <code>has_toddler</code>), or "household" when there's one group or none. The server accepts only these values
Symptoms	Wrapping grid of seven chips, multi-select: headache · congestion · cough · itchy or watery eyes · fatigue · trouble sleeping · wheezing or shortness of breath	<code>symptoms[]</code> — send stable keys, not display text: <code>headache</code> , <code>congestion</code> , <code>cough</code> , <code>itchy_eyes</code> , <code>fatigue</code> , <code>trouble_sleeping</code> , <code>wheezing</code> (so Spanish labels don't split the data)
Severity	Three-option control: Mild · Moderate · Bad — appears only after a symptom is chosen	<code>severity: mild / moderate / bad</code> , or <code>null</code> . Not the same scale as environmental severity

Illness flag	Checkbox: " <i>I think this might be a cold or flu, not the environment.</i> "	<code>possibly_illness</code> . The analysis drops these days entirely — which is the point: cold and flu season overlaps heating season (§14.5)
Note	Single line, expands; placeholder " <i>Anything else? (optional)</i> "; counter past 400; max 500	<code>note</code> (the server truncates at 500)
Save	Primary, full width: <i>Save</i> , or <i>Update</i> if an entry exists for this day and group. On save: brief toast, form collapses into the list with the new entry on top	Also <code>retrospective: false</code>
Felt fine today	Tertiary beneath: <i>Felt fine today</i> — saves an entry with no symptoms, toasts	<code>symptoms: []</code> , <code>severity: null</code> . This matters: without good days recorded, the analysis only ever sees bad ones (Tier 1, §65)
Save failure	" <i>Couldn't save that entry. It's still here — try again.</i> "	Form keeps everything

TRAP — EVERY SAVE REWRITES EVERY FIELD. Re-saving a day/group with a partial body resets whatever was left out (note → empty, symptoms → none, flags → false). Always send the whole entry, pre-filled from the existing one.

Recent entries list LOCKED

Below the form, up to seven recent entries. Each row: date, group as a small tag, symptoms as comma-separated display labels ("*Felt fine*" when empty; the retrospective key `unspecified` displays as **PROPOSED** "*Rough day*"), and that day's dominant condition as a severity pill. Note truncates to one line, full on tap (§47.4, §61).

BIND Day condition = worst of `history[date].air/uv/pollen/mold` (no_data if the date has no history). Offline entries show a pending marker until sent — only if the offline queue is built (Tier 3, A2); otherwise a failed save keeps the form filled and says so.

Delete: swipe left reveals delete; deletion is immediate (`DELETE /api/journal?id=`) with an undo toast — undo re-saves the same entry with `POST` (§47.4, §31.5).

Seasonal prompts — only on this page LOCKED

Seasonality appears in Journal and nowhere else (§14.7). As dismissible callouts at the top of Log mode, above the date header — never a modal, never interrupting the form. Dismissal lasts for that season (stored locally).

- Winter radon prompt** — October–February, household in radon zone 1 or 2: "*Radon usually peaks in winter in homes like yours. This is the most accurate time of year to test — a summer test can read low and miss the real number.*" Zone from the stored HomeGuard response (`radon.zone`); skip if unknown.
- Pollen season note** — e.g. "*It's tree pollen season in your area — elevated readings this month are normal.*" By month (NC: tree late winter–spring, grass into summer, ragweed in fall).
- Season summary** — e.g. "*This spring you logged 22 days. On the 9 highest-pollen days, you flagged symptoms 7 times.*" A milestone, not a nag.
BIND `GET /api/journal/summary` → `statement` , `logged_days` ; shown after a season ends when the household has entries from it (a `season_summary` alert also announces it).

Trends mode LOCKED

Element	Content / behavior	Binding
Calendar heat grid	The page's signature visual: one cell per day across recent weeks, colored by that day's worst severity, dot on days symptoms were logged. Answers Journal's whole question visually before any statistics — and shows gaps in logging without nagging about them	Cell = worst of <code>history[date]</code> factor words; no history for a date → no-data cell. Dot = any entry that date with non-empty <code>symptoms</code> . Today marked; future not drawn
Grid legend	Severity colors with labels + the dot labeled <i>you logged</i>	—
Day detail (tap a cell)	Compact sheet: date as title; every reading that day with its severity pill; what was logged, if anything; option to add or edit that day's entry	<code>history[date]</code> → air, uv, pollen, mold + values ; entries for that date
Findings	One card per finding: the statement, the underlying counts, a small pair of bars comparing the two rates	<code>findings[]</code> → <code>statement</code> , <code>elevated_rate</code> , <code>other_rate</code> , <code>counts.elevated_days</code> , <code>counts.other_days</code> , <code>factor</code>
Disclaimer	Permanent, never dismissible, under the findings	<code>disclaimer</code> — <i>"This is a pattern in your own logs, not a medical finding. Many things affect how you feel on a given day."</i>
Not ready	Replaces findings with progress: <i>"Keep logging — there isn't enough here yet to say anything reliable. You've logged 3 days; patterns need about 5."</i>	<code>ready === false</code> . The n is counted from <code>entries</code> : distinct dates with non-empty <code>symptoms</code> and <code>possibly_illness</code> <code>false</code> . Not <code>flaggedDays</code> — see below
Line chart	Environmental history over time with logged days marked; different marker per household group; horizontally scrollable	<code>history</code> + <code>entries[]</code> . <code>band</code>

DON'T SHOW `flaggedDays` **AS "DAYS LOGGED"** — the server's `flaggedDays` counts only symptom days that also have a day of readings and aren't flagged as illness — it's "comparable days," not "days logged." Count logged days from `entries` for the message. Findings need two things: 5 comparable symptom days (`flaggedDays` vs `needed.flagged`) **and** 14 days of readings (`totalDays` vs `needed.total`). When the household has logged enough but there aren't 14 days of readings yet (a new household before its history fills in), use **PROPOSED** *"HALO needs about two weeks of readings for your area to compare against. It has [totalDays] so far."*

MISSING COPY — READY BUT NO FINDINGS. When there's enough data but no factor shows a big enough difference, the server returns `ready: true`, `findings: []` . The blueprint has no copy for this. **PROPOSED** : *"No clear pattern yet. Your symptom days don't line up with any one condition so far."* — the disclaimer still shows.

WHY THE RESTRAINT IS THE FEATURE — findings are rates of co-occurrence, never a fitted model; illness-flagged days are excluded; factors with under three days on either side are skipped; differences under about 25 points aren't reported (§14.6). The frontend shows exactly what's returned and adds nothing. This is one of the strongest things to explain to a judge.

Page-level states

State	Shows
No entries at all	Log: the retrospective first visit. Trends: <i>"Nothing logged yet. Start by marking a few days that felt rough — we'll show you what was in the air."</i>
Loading	Trends: grid-shaped placeholder cells + two finding-card placeholders
Entries fail to load	Generic error + retry; the form still works for today
Offline	Default: the form stays filled and a failed save shows <i>"Couldn't save that entry. It's still here — try again."</i> (PROPOSED). With the Tier 3 queue built: entries queue with a pending marker (A2)

Connection to the Assistant LOCKED

Journal is the one feature that gives the Assistant information about the household itself: a question like "why does my son cough more on some days?" is answered from that household's own entries and environmental history, alongside cited guidance (§14.8). The server supplies this context to the Assistant; nothing for this page to do beyond saving entries — see C2.

Proposed content PROPOSED

Addition	From
Progress toward findings shown as two small counters in Trends — comparable symptom days 3 of 5, days of readings 9 of 14 — so it's clear which one is holding things up	<code>flaggedDays</code> , <code>totalDays</code> , <code>needed</code> (labeled as comparable days, not days logged)
In day detail, the household sentence for that day's worst reading isn't available from history — instead show the values with units	<code>history[date]</code> . <code>values</code>
Per-group filter on the heat grid dots when more than one group logs	<code>entries[]</code> . <code>band</code>

Visual decisions to make DESIGN

Heat grid (signature), segmented control, month calendar for the retrospective step, symptom chips (reusing toggle chips from onboarding), the three-option severity control, checkbox, toast (first use), empty state (first use), findings card with its bar pair, the scrollable line chart.

B5. Act

Purpose LOCKED

Everything a household can do beyond its own four walls — contributing, advocating, sharing. Distinct from Learn: Learn is personal protection (filters, sunscreen, closing windows); Act is acting on the problem itself. This is the layer that makes HALO civic rather than merely personal, and the clearest answer to why it belongs in a competition run by a member of Congress (§15.1, §17.2).

Address and entry

/act , Tab 5. Each of the four features opens a bottom sheet (history entries).

Shape LOCKED

Header *Act*. A framing line: *"Environmental problems aren't solved one household at a time. Here's what else you can do."* Then four full-width cards, each with one strong icon, a title, a one-line description, and a chevron. Styled editorially — taller cards, larger type, more whitespace, no charts — because these are invitations, not readings. Same card component; only proportions and type scale differ (§15.2, §48.1).

Data calls BIND

Call	Feeds
GET /api/profile	County, water source, renter mode
Stored (or fresh) GET /api/home-guard response	Which risks are elevated (volunteer causes, help sections), letter values, health card
Stored (or fresh) GET /api/daily-score response	Health card daily block
GET /api/volunteer?causes=...	Volunteer sheet
GET /api/district	PROPOSED district context card

Card 1 — Volunteer near you LOCKED

Description: *"Groups working on the problems HALO found in your area."* The list is filtered by this household's own findings — a household with elevated water sees the riverkeeper working on exactly that; one with clean water and high radon doesn't (§15.3).

Element	Content	Binding
Request	—	GET /api/volunteer?causes= — the frontend derives causes; the server doesn't. From HomeGuard: water item in action_plan → water,pfas ; radon item → radon ; lead item → water ; well/spring source → wells . Nothing else — adding a broad tag like advocacy would match every organization and undo the filtering. Nothing elevated → send no causes (every county-serving org is returned, unranked). County defaults to the profile's
Sheet title + basis line	Volunteer near you, then e.g. "Groups working on water quality in Union County."	county , causes from the response
Each organization	Name; description (2 lines max); cause tags; counties served (Statewide when "statewide"); Visit site opening externally; when it was last verified by a person	orgs[] → name , description , causes[] , counties[] , url , last_verified
Verification	last_verified set → "Last verified March 2026"; null → "Not yet verified"	Never claim verification that didn't happen
Footer	"Every group here was checked by hand. Last verified [month year]."	Show only if every listed org has last_verified — otherwise the claim would be false
Empty	"We don't have verified groups listed for your county yet."	orgs empty

NOTE — don't render match_count when no causes were sent (it's always 0 then). A household with nothing elevated sees every organization serving its county, unranked.

Card 2 — How you can help LOCKED

Description: *"Ways to act on what's affecting your area."* Guidance tied to this household's actual readings, never generic (§15.4). Sections appear only for risks actually elevated here; each has a heading naming the risk and two to four numbered actions, each one line with an optional supporting sentence. A closing *Always useful* section applies to everyone (§48.3).

Section appears when	Covers	Trigger binding
Radon zone is elevated	How to test, and why it matters beyond one home — radon is geological, so a high result is information the neighborhood should have too	radon.zone 1 or 2
Water contamination is elevated	Finding the utility's public comment periods; who to contact; what a Consumer Confidence Report is and how to read your own	action_plan has a water item
Nearby	How enforcement works, looking up a facility's record, filing a	A facility with open violations (open_violations > 0) within PROPOSED 5

facility violations	complaint	miles of the home, from <code>/api/map?layer=facilities</code> (distance from the profile's <code>lat / lng</code>)
Always	How to read a water quality report; what to ask at a town or utility meeting	—

CONTENT TO WRITE — the blueprint names each section's topic but not its actual numbered actions. Someone writes 2–4 actions per section, cited where they state a fact, following the voice rules. This is design/content-owner work (\$26.3b), not engineering.

Card 3 — Write a letter LOCKED

Description: "A *ready-to-send message about your water, filled in with your numbers.*" The single strongest fit for a congressional competition: a tool that helps a constituent engage an institution, built from their own measured data (\$15.5).

Element	Content / behavior	Binding
Recipient	Three-option control at the top: <i>Utility</i> · <i>Landlord</i> · <i>Representative</i> . Changing it regenerates the draft	—
Data notice	" <i>Filled in with your utility's most recent test results.</i> "	—
Draft	Editable multi-line text area filling most of a tall sheet	Utility template (verbatim below)
Actions	Primary <i>Copy</i> (toast confirms) · secondary <i>Share</i> (the device share sheet) · tertiary <i>Start over</i> restores the generated text after edits	<code>navigator.clipboard</code> , <code>navigator.share</code> (fall back to Copy where Share isn't supported)

I am a customer of [utility name], public water system [identifier]. Your most recent federal sampling, dated [date], reported [contaminant] at [value] parts per trillion. The federal maximum contaminant level is [limit] parts per trillion. This system serves approximately [population] people. I am writing to ask what your compliance timeline is, and what interim steps are available to households in the meantime.

Placeholder	Binding
utility name / identifier	<code>water.pws_name</code> / <code>profile.pwsid</code>
contaminant, value, limit, date	Worst enforceable entry of <code>water.scored_contaminants</code> → <code>contaminant</code> , <code>value_ppt</code> (1 decimal), <code>limit_ppt</code> , <code>date</code> as "July 2025"
population	<code>water.population_served</code> (formatted per A5); drop the sentence if null

CONTENT TO WRITE AND EDGE CASES — (1) The blueprint gives the utility letter only; the landlord version ("asks about testing and tenant remedies rather than compliance schedules") and the representative version need writing (content owner). (2) HALO doesn't store who the household's representative is; the letter opens "*Dear Representative,*" for the person to complete. (3) When there's no measured water result to cite (well, no data yet, lookup failed, nothing over a limit), show a short explanation instead of an empty template; for wells, hide the utility option.

Card 4 — Your home's health card LOCKED

Description: "A *one-page summary you can save or share.*" Its practical purpose is to be handed to someone: a doctor asking about exposure, a landlord being asked to test, a relative being persuaded to test their own home. Formats data every other page already produces, plus the source list (\$15.6, \$48.5).

Section	Content	Binding
Header	HALO mark, county, generation date	<code>profile.county</code> , <code>now</code>
Two scores	Daily and home scores side by side with their rings and labels	<code>score.display_score</code> from each stored response
Category table	Every category with its reading, severity, and the date behind it: air, UV, pollen, mold, water, radon, lead	Today readings + time retrieved; water worst contaminant + <code>latest_sample_iso</code> ; radon zone ("county estimate"); lead level + build year
Household notes	The household-aware sentences, if any	<code>*.sentence</code> , <code>lead.sentence</code>
Top actions	Top three action items	<code>action_plan[0..2]</code> (title, cost, certification)
Sources	Every source with the date it was retrieved	GET <code>/api/sources</code> → <code>name</code> , <code>last_retrieved</code>
Buttons	Primary <i>Save as image</i> · secondary <i>Print</i>	Client-side image render; print stylesheet

PRINT IS TIER 1 — printing must produce a clean black-on-white page with no interface chrome, no backgrounds, severity conveyed by label (and pattern) rather than color, and it's always light even in dark mode (\$48.5, \$59.3, \$65.1).

Page-level states

State	Shows
HomeGuard not yet loaded this session	Cards still render; each sheet fetches what it needs on open with its own skeleton
Loading (page)	Four card-shaped placeholders (the page itself needs almost no data)

Proposed content PROPOSED

Addition	From	Why
A district context card above the four features: e.g. " <i>[affected_population] people in NC-08 are served by a water system over a federal PFAS limit.</i> " ("at least" when <code>affected_population_complete</code> is false). Tap → Map's district panel	GET <code>/api/district</code> → <code>affected_population</code> , <code>affected_population_complete</code> ,	Puts the household's result in civic context — "is this just my problem?" — on the civic page. Strongest single line for a judge

	assembled_at	
In the letter sheet, a line showing which reading the letter cites, with a link to it in HomeGuard	The chosen scored_contaminants entry	Transparency: the person sees exactly what they're sending

Visual decisions to make DESIGN

The editorial card treatment (taller, larger type, one strong icon each), organization list rows, the letter editor sheet, the health card as a formatted document (on screen and in print), the district context card.

B6. Settings

Purpose LOCKED

Everything entered during onboarding must be changeable, and everything HALO knows must be inspectable. A household that can't correct a wrong address or remove a group will stop trusting the results (§16.1).

Address and entry

`/settings`, from the gear on any tabbed page. Header *Settings* with a back arrow (not a gear). No tab highlighted. Eight accordion sections (seven until reviewed Spanish exists — the Language section is hidden until then); only one open at a time; each row shows its current value when closed. Every control saves immediately and toasts only on failure (§54). Which section is open isn't kept in the URL.

Data calls BIND

Call	Feeds
GET <code>/api/profile</code>	<code>profile.county</code> , <code>pwsid</code> , <code>water_source</code> , <code>home_year</code> , <code>renter_mode</code> , <code>locale</code> , <code>household</code> , <code>household_set</code>
PUT <code>/api/household</code>	Household groups, renting/owning, language
PATCH <code>/api/profile</code>	Build year, water source
POST <code>/api/onboard</code>	Change address (via the onboarding address step)
GET / PUT <code>/api/notifications</code> , POST / DELETE <code>/api/push/subscribe</code>	Notification switches and device notifications
DELETE <code>/api/journal?all=true</code>	Clear journal
DELETE <code>/api/account</code>	Delete everything
GET <code>/api/sources</code>	About → data sources and dates

ONE SAVE FUNCTION FOR `/api/household` — it replaces all seven groups on every call — send only `renter_mode` and the household's groups are wiped. Every Settings save that touches this endpoint goes through one function that sends all seven current booleans plus whatever changed. The PUT's response only echoes `renter_mode` / `locale` if you sent them, so keep local state from the profile, not from the PUT response.

The eight sections LOCKED

Section	Collapsed summary	Expanded	Binding
Household	" <i>Toddler, adult</i> " or " <i>Not set</i> " — "Not set" whenever no group is true. Don't rely on <code>household_set</code> alone: saving renting or language through PUT <code>/api/household</code> also sets it to true	The same seven chips as onboarding, same order; each tap saves; takes effect on the next reading refresh. (The blueprint conflicts with itself: §49.2 says toast on save, §54 says toast only on failure. Pick one for all of Settings — failure-only is quieter)	Summary from <code>household</code> + <code>household_set</code> . Save: PUT <code>/api/household</code> with all seven. Chip flips instantly, reverts on failure
Your home	" <i>Union County · City water</i> "	Build year field (A7 rules), water source dropdown (same five options as onboarding — "Not sure" is saved as <code>other</code> , so a stored <code>other</code> preselects "Other"), and secondary <i>Change address</i>	Build year and water source: PATCH <code>/api/profile</code> , saved on change (build year on blur). Change address → confirm " <i>This will replace your current results. Continue?</i> " → onboarding's address step; re-geocodes and discards the text again, exactly like onboarding. Clear stored Today/HomeGuard readings afterward, then request GET <code>/api/daily-score?fresh=1</code> — the server's one-hour cache doesn't know the location changed and would otherwise keep serving the old address's readings (on a 429, keep the stored-readings notice and retry after <code>retry_after_seconds</code>) — then GET <code>/api/home-guard</code> . PROPOSED flow: after the address step, continue to step 4 (Your home) prefilled with the current build year and water source — a new home may have a different water source — then the results reveal, then Today. Household (step 3) is skipped. Yogi's call
Renting or owning	<i>Owner / Renter</i>	Two-option control + one sentence that it changes which actions are recommended everywhere	<code>renter_mode</code> via PUT <code>/api/household</code> (with all seven groups). Changes every action recommendation (§12.7)
Language	<i>English</i>	English · Español, applying immediately	<code>locale</code> via PUT. Interface strings switch from the translation layer; Learn content is requested with the new <code>locale</code>
Accessibility	<i>Default</i>	Text size slider with live preview text; higher-contrast switch; reduce-motion switch	Device-local (stored in the browser) — these are about this device, not the household. Reduce motion defaults to the system setting
Notifications	" <i>4 of 5 on</i> "	One switch per alert type with its plain-language	Types: GET / PUT <code>/api/notifications</code> → <code>preferences.air_quality_change</code> , <code>weather_advisory</code> , <code>new_water_results</code> , <code>radon_season</code> , <code>season_summary</code> . Device

		name and one-line description, all default on; plus a switch for device notifications on this browser	notifications: request browser permission only when the person turns this on (and only after they've used the app at least once, \$19.5), then <code>POST /api/push/subscribe</code> ; turning it off → <code>DELETE</code>
Data and privacy	—	The full privacy statement (verbatim below), then destructive <i>Delete everything</i>	<code>DELETE /api/account</code>
About	"Version 1.0"	Every data source with the date it was last retrieved; how scores are calculated; who built HALO; link to the source repository	<code>GET /api/sources</code> + static text (below)

Notification switch copy PROPOSED

Preference	Name	Description
<code>air_quality_change</code>	Air quality changes	When today's air gets worse than yesterday's. Earlier for homes with someone sensitive
<code>weather_advisory</code>	Weather advisories	Flood, boil-water, and heat advisories for your county
<code>new_water_results</code>	New water results	When new federal testing results are published for your utility
<code>radon_season</code>	Radon testing season	A winter reminder when testing is most accurate
<code>season_summary</code>	Season summaries	A look back at your journal when a season ends

Privacy statement — verbatim LOCKED

What HALO stores: your approximate coordinates, your county, which water utility serves you, your home's build year, and which household groups you told us about.

What HALO never stores: your street address, anyone's name, anyone's age or birthday, and any medical record. Your journal notes stay on your account and are never shared with anyone.

When you ask HALO's assistant a question, the readings and logged symptoms that help answer it are sent to its AI service to write the answer. Your notes are never sent.

Also stated here, per §37.4: HALO collects no behavioral analytics.

OPEN — THE "STORES" LINE IS INCOMPLETE — §16.2 asks for exactly what is and isn't stored. The locked line omits water source, renting or owning, language, state, daily reading history, journal entries, and alerts. PROPOSED addition for Yogi to approve (don't change the locked text without it): *"It also stores your water source, whether you rent or own, your language, your daily readings, the journal entries you write, and the alerts it sends you."*

DECIDED — THE ASSISTANT AND JOURNAL DATA — to answer questions like "why does my son cough more on some days?" (§14.8, §18.2), the Assistant sends the relevant readings and journal symptom names/dates to its AI provider — never the free-text notes. Team decision: acceptable for this build, since HALO isn't shipping as a published product handling real household data at scale. The privacy statement above reflects this; "never shared" now scopes to notes only, not the symptom log as a whole.

Delete everything LOCKED BIND

Tapping *Delete everything* opens a compact sheet stating exactly what gets deleted — *"This permanently deletes everything HALO knows about your household. This cannot be undone."* — with a field that requires typing *DELETE*, and a destructive confirm button that stays disabled until it matches (§49.3). Confirm → `DELETE /api/account` with body `{ "confirm": "DELETE" }`. On `{ deleted: true }`: clear every locally stored reading, flag, and preference, discard the session, and return to onboarding with a brief confirmation. On failure: stay on the sheet with the generic error — nothing was deleted. Tier 1 per the blueprint's own triage.

About — content LOCKED

- Every data source named individually, each with when its data was last retrieved (§16.4) — from `/api/sources`: federal water contaminant testing (EPA UCMR 5), the federal air quality network (AirNow), the modeled weather and air service used as fallback (Open-Meteo), the pollen service (Google Pollen), the national weather service used for the mold estimate (NWS), county radon zone data (EPA), the water-system boundary and population data (EPA), and facility compliance records (EPA).
- How scores are produced, in plain language: daily conditions combine as independent risks; home conditions combine multiplicatively; missing data is excluded rather than counted as zero; one severe problem can't be averaged away by several mild ones.
- Who built it; link to the repository.

Language and Spanish

DECISION — the blueprint's rule: machine-translated health guidance isn't acceptable; ship English-only rather than unreviewed Spanish (§68). The Language section appears only once a person has reviewed the Spanish interface strings and Learn content. Either way, every string goes through a translation layer from day one, no button's width is fixed to its English label, and Learn trusts the `locale` it gets back (it reports the language actually served).

Proposed content PROPOSED

Addition	From
"What HALO knows about you" at the top of Data and privacy — the actual stored values shown back: county, utility ID, water source, build year, groups	<code>GET /api/profile</code> . Makes "everything must be inspectable" (§16.1) literal

Clear journal in Data and privacy, confirmed with "*This deletes all [n] of your entries permanently.*"

DELETE /api/journal?all=true . The confirmation copy exists (\$31.5) but the blueprint never places the control

Visual decisions to make DESIGN

Accordion rows with value summaries, switches, the text-size slider with live preview, the destructive button and delete sheet, the formatted privacy statement.

B7. Map and the District panel

Purpose LOCKED

Answer *is this just my problem, or is it everyone's?* Every other page is about one household; this one places it in its region — what turns a personal utility into something civic. The most visually striking screen, so it matters disproportionately in a demo (§13.1).

Address and entry

`/map` (Water layer, overall severity), Tab 3. `?layer=radon|facilities|air`, `?layer=water&contaminant=pfos`, `?layer=water&quarter=2024Q3`, `?system=<pwsid>` restore state. The header sits over the map. On wide screens the map column widens to 600px — the one exception to the 430px cap (§53.2). The map library loads only when this tab is first opened, never on app start (§35.2).

Data calls BIND

Call	Feeds
<code>GET /api/profile</code>	Home pin (<code>lat</code> , <code>lng</code>), own system (<code>pwsid</code>), <code>county</code>
<code>GET /api/map?layer=water</code> (<code>&quarter=</code> when scrubbing)	<code>features[]</code> , <code>counts</code> (the live count line), <code>quarters[]</code> , <code>assembled_at</code>
<code>GET /api/map?layer=radon</code>	<code>features[]</code> → <code>county</code> , <code>zone</code> , <code>severity</code> ; joined to bundled NC county shapes by county name — the API's names include the suffix (<code>"Cabarrus County"</code>) while most shape files don't, so normalize before joining
<code>GET /api/map?layer=facilities</code>	<code>features[]</code> → facility points
<code>GET /api/map?layer=air</code>	<code>features[]</code> → monitoring stations
<code>GET /api/district</code>	Comparison banner, district panel (NC-08)
Static asset bundled with the app	NC county boundary shapes for the radon fill

Each layer is fetched the first time it's selected and cached for the session (and offline, §36). Map data is built once a day on the server, so every layer loads as one fast request.

Layout LOCKED

The map fills everything between header and tab bar, edge to edge. Floating on top: a row of layer chips; beneath them (Water only) the contaminant selector, a live count line, and a conditional rescission notice; a locate button (crosshair icon — tapping it animates to the home pin at the default zoom, §54) and zoom buttons on the right edge (the blueprint hides zoom buttons on touch devices; that's Tier 3 — **default: visible everywhere**, §65.3); a legend lower left; the time slider (Water only, when the data has quarters); a comparison banner; and a handle at the very bottom, *"See the full NC-08 picture,"* opening the district panel (§13.2, §46.2).

Initial state and memory LOCKED

Centered on the household at a zoom showing roughly their county and its neighbors — close enough to identify their own system, wide enough to show others exist. Home pin visible; Water layer, overall severity. On return within a session, restore the last viewport, layer, and selection (§46.3).

Layer chips LOCKED

Water · Radon · Facilities · Air. One active; active chip visibly filled. Switching re-renders data without moving the viewport, so the household's position stays anchored (§13.3). Arrow keys move between chips.

What each layer draws LOCKED BIND

Layer	Drawn as	Binding
Water	One circle per public water system at its service-area center. Size by population served (square-root scale so big systems don't swamp small ones); fill by severity. No reading for the selected compound → small grey marker, legend PROPOSED <i>"No detection reported"</i> — not "Not tested for this": the loaded EPA data kept only detections (non-detects were dropped), so a system with no reading may have been tested and found nothing. The same limit affects the district's <code>tested</code> counts until Vibhav reloads the data with non-detects or adds a per-compound tested flag. Water and air layers never cluster (§46.5) — the pattern is the point. Systems without a location are listed in the district panel instead of drawn	<code>lat</code> , <code>lng</code> , <code>population</code> , <code>overall_severity</code> or the selected compound's <code>severity</code> . <code>located</code> vs <code>count</code> tells you how many can be drawn
Radon	Filled county shapes by the county's radon-zone severity, low enough fill opacity that county names stay legible	<code>features[]</code> . <code>severity</code> , <code>zone</code>
Facilities	Points colored by compliance; clusters that separate as you zoom in; cluster badge shows a count; tapping a cluster zooms to fit it rather than opening a sheet (§46.5)	<code>lat</code> , <code>lng</code> , <code>severity</code> , <code>compliance_status</code> (<code>no_violation</code> · <code>violation</code> · <code>significant_violation</code> · <code>unknown</code>)
Air	One point per monitoring station by current air level, visually distinct from water circles	<code>lat</code> , <code>lng</code> , <code>severity</code> , <code>aqi</code>
Home pin	Always on top of every layer, never recolored by severity — the household never loses itself	Profile <code>lat</code> / <code>lng</code>

The contaminant selector (Water) LOCKED

The single most interactive element in the app — it lets someone answer their own questions rather than read a fixed answer (§13.4). Choosing a

compound recolors every system by that compound alone, against its own limit, animated over about 300ms.

Option	Colors by	Binding
Overall water health (default)	Each system's combined severity across everything measured	<code>overall_severity</code>
PFOA · PFOS	That compound vs its 4 ppt limit	<code>contaminants.PFOA.severity</code> , <code>contaminants.PFOS.severity</code>
PFHxS · PFNA · GenX	vs its 10 ppt limit, with the rescission notice	<code>contaminants.PFHxS</code> , <code>.PFNA</code> , <code>["HFPO-DA"]</code> ; notice when <code>proposed_for_rescission</code>
Combined hazard index	The federal combined measure — 1.0 or above is a violation even when no single compound exceeds its own limit	<code>hazard_index.severity</code> (<code>value</code> shown in the sheet), with the rescission notice when <code>hazard_index.proposed_for_rescission</code>

- An untested entry is `{ severity: "no_data", limit_ppt }` with no value, ratio, or date — check before reading them. Absence of a measurement is never shown as safety.
- `overall_severity: "no_data"` with `coverage: "unscorable"` means compounds were detected but none has a federal limit — legend entry **PROPOSED** *"Detected — no federal limit to compare."*
- Live count beneath the selector, updating with the selection — the most quotable line on the page: *"38 of 412 systems exceed the limit for PFOS."* (§13.4). Bind the map response's own `counts.by_contaminant[X].over` and `.tested` (and `counts.over_limit` of `counts.total` for overall, `counts.hazard_index_over` for the hazard index) — they describe exactly the systems drawn, statewide. The district panel's figures are NC-08 only.
- Rescission-flagged selection (a compound's `proposed_for_rescission` , or `hazard_index.proposed_for_rescission` for the hazard index) → one-line notice beneath the count that the standard itself is contested.
- The household's own system (`pwsid` matches the profile) stays outlined and labeled regardless of selection.
- Any other layer active → the selector is replaced by that layer's legend (radon zones, facility compliance, air quality levels).

Tapping a feature LOCKED

Every feature opens a compact bottom sheet over the map (never navigates away); the feature shows as selected while its sheet is open and deselects when it closes; the blueprint pans the map just enough to keep it visible above the sheet (§13.6, §46.6–46.7) — that's Tier 3: **default is no auto-pan**; the sheet may cover part of the map and people drag (§65.3). Each sheet: title row, severity pill, body, footer with the data date.

Feature	Sheet contents, in order	Binding
Water system	System name (title); ID beneath; severity pill; population served; threshold bar chart of its contaminants (untested ones listed as "Not tested"); hazard index; sample date; <i>Learn about PFAS</i>	<code>name</code> , <code>pwsid</code> , <code>overall_severity</code> , <code>population</code> , <code>contaminants</code> , <code>hazard_index</code> , <code>latest_sample_iso</code>
County (radon)	County name; zone as a severity pill; two sentences on what the zone means; the sentence that zones predict county averages rather than measuring a home; <i>Learn about radon</i>	<code>county</code> , <code>zone</code> , <code>severity</code>
Facility	Facility name; compliance status pill; which programs regulate it (small tags); violations in the last 3 years; penalties in the last 5 years (dollars) if any; last inspection date	<code>name</code> , <code>compliance_status</code> , <code>severity</code> , <code>programs[]</code> , <code>recent_violations</code> , <code>penalties</code> , <code>last_inspection</code>
Air monitor	Station name; current index and severity pill; dominant pollutant; distance from the household; last reading time	<code>name</code> , <code>aqi</code> , <code>severity</code> , <code>dominant_pollutant</code> , <code>last_reading_at</code> ; distance computed from the profile location
Home pin	<i>Your home</i> ; county; today's score; primary <i>Go to Today</i>	<code>profile.county</code> , stored daily <code>display_score</code>

Time slider (Water) LOCKED

Federal water testing is quarterly, so the water layer can move through time: a horizontal slider with quarter labels above the comparison banner; dragging recolors every system to its severity at that quarter, showing whether contamination is spreading, receding, or steady (§13.7). **BIND** ticks from `quarters[]` ; each position requests `?player=water&quarter=` . **Shown only when `quarters` has two or more entries** — per the blueprint, if the data holds only the latest reading, the slider is cut and the layer is static (§68).

Comparison banner LOCKED

A small floating card near the bottom with one sharp, specific comparison, updating with the layer; tap → district panel (§13.8). Deliberately a single number rather than a chart — specific rankings are what people repeat to each other.

Layer	Banner	Binding
Water	<i>"Cabarrus County ranks 34th of 100 North Carolina counties for water quality."</i> (§13.8)	<code>district.household_county</code> → <code>county</code> , <code>water_rank</code> , of
Radon	PROPOSED <i>"[County] is Zone [n]. [k] of 100 North Carolina counties are Zone 1."</i>	Profile county + radon features
Facilities / Air	PROPOSED e.g. <i>"[n] facilities within 10 miles have open violations."</i> · <i>"The nearest air monitor is [d] miles away."</i>	Layer features + profile location

Freshness LOCKED

Every layer shows when its data was assembled, e.g. *"Updated today at 4:00 AM"* — undated environmental data invites the reasonable suspicion that it's stale (§13.9). **BIND** `assembled_at` .

The district panel LOCKED

An expandable panel inside Map — not a separate page. Opened by dragging the handle up or tapping the comparison banner. Shows no household data — it's the view a community group or legislative office could open (§20).

Content, in order	Binding
Headline figure: people	<code>affected_population</code> ("at least" when <code>affected_population_complete</code> is false), <code>systems_over_limit</code>

affected across the district, at hero size, with a one-line caption	
Comparison bar pair: district vs state (share of systems over a limit)	<code>state_comparison.district_share_over</code> , <code>.state_share_over</code>
County table: one row per NC-08 county — name, worst water severity, radon zone	<code>counties[]</code> → <code>county</code> , <code>worst_water_severity</code> , <code>radon_zone</code> , <code>radon_severity</code> , <code>systems_over_limit</code>
Per-compound detail: how many systems tested, how many over, how many people served by those	<code>by_contaminant[X]</code> → <code>tested</code> , <code>over</code> , <code>population_over</code>
Named local specifics: the Concord treatment plant serving 120,000+ customers above the federal limit; Union County's reading against the federal maximum; compliance timelines extending to 2031	Written content with sources and dates — every figure verified before launch (§66)
Every figure shows the date its data was assembled	<code>assembled_at</code>

Loading and failure LOCKED

- Loading: neutral canvas with a centered spinner and "*Loading map data.*" Controls present but disabled (the one place a spinner is right, §46.9).
- Layer data unavailable: base map + home pin + a callout that the layer's data couldn't load, with retry.
- Map library fails entirely: show the district panel's content as a full page, so the tab is never empty.
- No data in view: "*No data for this layer in the area you're viewing. Try zooming out.*"
- Offline: last loaded datasets, cached after first load (§36).

Proposed content PROPOSED

Addition	From
In the household's own water-system sheet, a line comparing it to the district: "Your system is one of [n] in NC-08 over a federal limit"	<code>district.systems_over_limit</code> , own system's severity
A "systems not shown on the map" count in the district panel, with their names and severities as a list	<code>count - located</code> , features without coordinates

Decisions to make

Technical: which map library (Mapbox GL JS on the existing Mapbox account, or MapLibre with free tiles); the source and file size of the NC county shapes (first-load budget under 500 KB, §35.1 — the map loads separately). Performance targets: first render under 2s, layer switch under 300ms (§35.1). **DESIGN** : a muted, low-saturation base map where the data layer is the only saturated thing on screen (§57.1's principle), switching to a dark variant with the same muted treatment in dark mode (§59.3); floating control styling; chip row; legend; home pin; cluster badges; district panel.

Part C — Overlays

All three are bottom sheets over the current page: they push a history entry, trap focus, lock the page's scroll, dismiss by drag / backdrop / Escape / close control, and only one is ever open (A1).

C1. Learn

Purpose and the rule that governs it LOCKED

Explain what a number means, at the moment someone is looking at it. Never reachable from the tab bar, no index, no browsable encyclopedia — content that can only be reached from a household's own reading, and references that reading in its second sentence, can't be generic by construction (§17.1).

Opened from

Where	Topic	Context params to send BIND
Today — Air card	air	source = air.source , value = air.aqi , severity = air.severity
Today — UV / Pollen / Mold cards	uv / pollen / mold	value and severity of that reading (pollen: the dominant category's value)
HomeGuard — Water card, threshold bar row, map water sheet	pfas	contaminant , value , limit (the tapped row, or the worst)
HomeGuard — Radon card, map county sheet	radon	county , zone
HomeGuard — Lead card	lead	home_year (omit → the "we don't know when this home was built" text)

Call: GET /api/learn?topic=<topic>&locale=<locale>&...context . URL form: ?learn=<topic> on the opening page. Cache each topic after first read (§36).

Anatomy, top to bottom LOCKED BIND

Element	Content	Binding
Title	Topic name	title
Your reading	The household's own reading as a severity pill with its value, right under the title — what makes this contextual rather than generic	Passed in from the page that opened it (not from this endpoint)
1. What it is	Two sentences, plain language	what_it_is
2. Why yours reads this way	References the household's actual value and situation	why_yours
3. What it means for your household	Branches on household composition	household_note — section omitted when null
4. How to protect yourself	A short checklist, most effective first; every item a concrete, checkable step; certifications named inline so they can be read standing in a store aisle. The section that makes Learn worth opening twice	protect[]
5. Sources	Every claim's origin, each with the date retrieved, each a working link, visually distinct	sources[] → label , url , retrieved

LANGUAGE — the response's locale is the language actually served. If Spanish was requested but a topic hasn't been translated and reviewed yet, it comes back in English with locale: "en" — show it, don't pretend otherwise.

THE LEARN / ACT BOUNDARY — Learn is personal protection (sunscreen, filters, flushing taps, closing windows, humidity). Act is collective action (volunteering, advocacy, sharing). Keeping that line clean keeps both coherent (§17.2).

States: skeleton (title bar, a pill shape, five paragraph blocks); error → generic error + retry. **DESIGN** : typographic treatment — strong headings, key facts given weight, sources as visible footnotes rather than hidden links (§6.7).

C2. Assistant

Purpose LOCKED

Answer questions about the household's own readings and about environmental health generally, grounded entirely in cited sources, without ever practicing medicine (§18.1). It draws on three things only: the household's current readings, the household's symptom entries, and a fixed curated set of source documents — never the web, never general knowledge (§18.2).

HOW IT KNOWS THE HOUSEHOLD — the server assembles the household's context itself, from the verified session — today's readings, the home assessment, and journal symptoms and findings when they exist (symptom names and dates only — **never the free-text notes**, per the Sept 27 privacy decision, B6) — and weighs it by the page the question came from. The frontend sends only `{ question, page }`; it never sends household data in the request. When an answer uses that context, the response says so (`uses_household_data: true`).

Entry LOCKED

The floating button on all five tabbed pages. Opens as the tall sheet. Carries the current page as context. The conversation isn't kept in the URL.

Calls BIND

Call	Sends	Returns
<code>GET /api/assistant?page=...</code> (on open)	<code>page</code> : Today → <code>today</code> , HomeGuard → <code>homeguard</code> , Journal → <code>journal</code> , Map → <code>map</code> , Act → none	<code>suggestions</code> (exactly 3), <code>disclaimer</code>
<code>POST /api/assistant</code>	<code>{ question, page }</code> (question trimmed, 1–1000 characters; <code>page</code> as for GET)	One of the branches below

Anatomy LOCKED

Element	Content / behavior
Title row	<i>Ask HALO</i> + close control
Suggested questions	On first open only: three tappable chips chosen by the page it came from — from HomeGuard, water questions; from Today, air questions (§18.6). Tap sends it
Conversation	The person's messages on one side, answers on the other. Answers never truncate
Thinking state	An in-progress indicator in the answer position until the response arrives. Answers arrive complete, not streamed; the indicator carries the wait
Citations	Under each answer: each cited source as a tappable name that opens it externally. Every answer carries at least one citation — an answer that can't be cited is not sent
"Uses your readings" marker	A small icon and the label <i>"Uses your readings"</i> above answers that used household data (§50.2) — <code>uses_household_data === true</code>
Declined answer	Shown in a caution callout, visually distinct from a normal answer
Disclaimer	Fixed above the input, always: <i>"Not medical advice. Talk to a clinician about health decisions."</i>
Input	Expanding text field + send button. The blueprint conflicts: §50.2 says send is disabled while empty; §34.1 says submit is never disabled for validation. Recommendation: follow §34.1 — send is always enabled; an empty or whitespace-only question (trimmed client-side) shows <i>"Ask a question to get started."</i> ; over 1000 characters → <i>"That question is too long."</i> Send is disabled only while a question is in flight

Response branches — how to tell them apart BIND

Check in this order:

Condition	Meaning	Show		
HTTP 429	Rate limited (20/hour per household; 300/hour app-wide)	<code>message</code> : "You've reached the question limit for now. Please try again in a little while."		
<code>declined === true</code>	Diagnostic refusal — a check that runs before any answer is composed	<code>message</code> in the caution callout		
<code>configured === false</code>	The assistant isn't configured on this server (e.g. a development environment without a model key)	PROPOSED <i>"The assistant isn't available right now."</i> — not the server's <code>message</code> , which reads as "no source for that question" and would mislead		
<code>typeof answer === "string"</code>	Grounded answer	<code>answer</code> <code>text</code> + <code>citations[]</code> → <code>{ n, label, url, retrieved }</code> + <code>uses_household_data</code> . The <code>[n]</code> markers in the text match each citation's <code>n</code> ; they aren't guaranteed to be in order or consecutive		
<code>answer === null</code>	No answer	Show <code>message</code> . <code>reason</code> : "no_source" → no source covers it (<i>"I don't have a reliable source for that. I can only answer from the health and environmental agencies I've been given."</i> , §31.3); <code>reason</code> :	No source found, or	<code>message</code> : no-source → <i>"I don't have a reliable source for that. I can only answer from the health and environmental agencies I've been given."</i> ; infrastructure →

		"unavailable" → couldn't reach sources right now, add Retry	couldn't reach sources	"I couldn't reach my sources just now..." + retry
--	--	---	------------------------	---

Required behavior LOCKED

- Household data and general guidance always distinguished in wording — "your reading today is 68" and "federal guidance suggests..." are separate statements (§18.4).
- Diagnosis questions declined before any answer exists, reliably — a demo audience may try (§18.4). The refusal is code, not an instruction.
- No source → says so rather than improvising.
- Answers can take up to about 20 seconds (the blueprint's target model timeout, §62.2 — not enforced server-side yet): the thinking state holds that long, and the client timeout for `POST /api/assistant` is 25s, not the 8s default. A client timeout is treated like `reason: "unavailable"` — the "couldn't reach" message with Retry.
- The Assistant is the last build and the first cut (§26.3, §68) — build its entry point so it can be switched off without leaving a dead button.

WHAT THE REFUSAL COVERS — questions asking for a diagnosis, whether a symptom means an illness, or what treatment or medication to take. Questions about how serious a *reading* is ("Is an AQI of 160 serious?") or how to fix an environmental problem ("What are the treatment options for radon?") are environmental questions and get answered.

BEFORE LAUNCH — federal guidance on wellness and decision-support software was revised in January 2026; how the assistant is described publicly should be checked against the current text (§18.5, §66). Test it against ten deliberately adversarial questions before any demo (§66).

DESIGN : message bubbles, citation chips, the thinking indicator, the declined-answer callout, the input row.

C3. Alerts

Purpose LOCKED

Tell a household when something meaningfully changes, without them opening the app to find out (§19.1). Opened only from the bell on Today's header.

Calls BIND

Call	Use
GET /api/alerts	count , unread (badge), alerts[] newest first (max 100)
POST /api/alerts { id }	Mark one read (it's POST, not PATCH)
POST /api/alerts { all: true }	Mark all read

The inbox LOCKED

Element	Content / behavior	Binding
Title row	Alerts + tertiary <i>Mark all read</i>	{ all: true }
Row	Reverse chronological. Unread: small colored marker + bolder title. Title, one-line message, relative time	title , message , fired_at , read ; marker color from severity when set (air alerts only), neutral otherwise
Tap	Expands to the full message and marks it read	POST { id }
Go to the related page	The blueprint conflicts: §19.4 says tapping an expanded row again navigates; §50.3 puts a tertiary button inside the expanded row instead. Recommendation: the button (§50.3) — a second tap on the same row is easy to trigger by accident	By type: air_quality_change → Today · radon_season → /home?risk=radon · new_water_results → /home?risk=water · weather_advisory → Today · season_summary → /journal?mode=trends · unknown type → no navigation
Swipe	Dismiss, immediate, with an undo toast (§19.4, §50.3)	POST { id, dismissed: true } ; undo → POST { id, dismissed: false }
Empty	"No alerts yet. We'll let you know when something changes."	alerts empty

History is kept rather than cleared, so the alert system's behavior can be seen without waiting for a live trigger — useful for a household back from a week away and for anyone evaluating the app (§19.4).

What generates alerts LOCKED

	Fires when	Example message (§19.2)	Opens
air_quality_change	Today's air is a worse level than yesterday's — one level earlier for households with someone with a respiratory condition, a toddler, or a senior (§19.3)	"Air quality dropped to Unhealthy for Sensitive Groups today."	Today
weather_advisory	A flood, boil-water, or heat advisory is issued for the county	"Flood advisory for Union County. Flooding can affect well water — retest after it recedes."	Today
new_water_results	New federal testing results change the household's utility readings	"New water testing results were published for your utility."	HomeGuard, water card
radon_season	October-February, household in an elevated radon zone — once per winter	"Winter is the most accurate time to test for radon."	HomeGuard, radon card
season_summary	A season ends and the household has journal entries from it	"Your spring summary is ready."	Journal, Trends

Types the household switched off in Settings are never generated (/api/notifications). Show any unrecognized type with its title and message and no navigation.

Device notifications LOCKED

Use the browser's notification capability where permission was granted; otherwise alerts simply appear in the inbox on next open. Ask permission only after the household has used the app at least once, never on first load (§19.5) — in practice, from the device-notifications switch in Settings. Subscribing registers the browser with POST /api/push/subscribe ; a notification tap opens the related page. Requires a service worker (which also serves the offline behavior, §36).

DESIGN : row anatomy, unread marker, the bell badge.

Part D — API Reference (the finished backend)

Every endpoint and field this document binds to, in one place. This is the contract between the frontend and the finished backend: the backend punch list uses exactly these names. Field names are verbatim — bind to them as written.

D0. Conventions that apply to every endpoint

Rule	Detail
Identity	Every route except the daily job requires <code>Authorization: Bearer <session token></code> . No route accepts identity from a body or query parameter.
401 vs 503	<code>401</code> = the session is definitively invalid (the account no longer exists) → the client discards the session and starts a new one, silently. <code>503</code> = identity couldn't be verified right now → the client retries and never resets the session. This distinction is what protects a household's data during an auth-service outage.
Errors	<code>{ "error": string }</code> . Never shown to people verbatim — mapped to the copy in A2. Exceptions are listed per endpoint (onboarding's address errors, the assistant's messages).
Absent fields	Any field can legitimately be <code>null</code> or missing, and the interface handles that explicitly — an em dash, a No data state, a hidden element — never a crash and never a zero (§62.4). This is also what lets a page be built before every field exists.
Severity	A closed set: <code>good</code> , <code>moderate</code> , <code>elevated</code> , <code>high</code> , <code>severe</code> , <code>no_data</code> . Anything else renders as No data. Always read the field named <code>severity</code> (or <code>level</code> for lead). Some objects also carry an older <code>status</code> field kept for compatibility — never bind it.
Dates	Daily readings and journal entries: <code>YYYY-MM-DD</code> in the household's local date. Timestamps: ISO 8601. Water sample dates: bind the <code>*_iso</code> fields (<code>YYYY-MM-DD</code>); the older <code>M/D/YYYY</code> strings remain for compatibility only.
Freshness	Every cached or pre-built response states when its data was retrieved or assembled (<code>retrieved_at</code> , <code>assembled_at</code>).
Change discipline	Additive only once frontend work begins: fields may be added, never renamed, removed, or re-typed (§62.4).

D1. Household and account

Endpoint	Request	Response
<code>GET /api/profile</code>	—	<code>{ onboarded, onboarding_complete, profile: { county, state, pwsid, lat, lng, home_year, water_source, renter_mode, locale }, household: { has_toddler, has_child, has_teen, has_adult, has_senior, has_pregnant, has_respiratory }, household_set }</code> . <code>onboarding_complete</code> is true once coordinates <i>and</i> water source are stored — route on this, not <code>onboarded</code> . A database failure returns 500, never an empty profile.
<code>PATCH /api/profile</code>	<code>{ water_source?, home_year? }</code>	Same shape as GET. <code>home_year</code> : integer 1700-current year or <code>null</code> . <code>water_source</code> : <code>utility</code> · <code>well</code> · <code>spring</code> · <code>other</code> (the onboarding labels are also accepted and normalized)
<code>POST /api/onboard</code>	<code>{ address }</code> — may be "lng,lat" from device location. (Water source and build year are saved with <code>PATCH /api/profile</code>)	<code>{ profile_id, lat, lng, county, state, pwsid, service_area_status, water_source, home_year }</code> . <code>state</code> is the two-letter code, e.g. "NC". <code>lat</code> / <code>lng</code> are rounded to about 100 m before storage. The response also echoes <code>zip</code> , which is no longer stored — don't bind it. <code>service_area_status</code> : <code>measured</code> (a utility boundary contains the point) · <code>outside_known_area</code> (none does) · <code>lookup_failed</code> (the boundary lookup failed; a previously stored utility is kept). Errors: 400 no address · 404 not found · 429 too many lookups · 500
<code>GET /api/household</code>	—	<code>{ household: {7 booleans}, household_set, renter_mode, locale }</code>
<code>PUT /api/household</code>	<code>{ has_toddler ..., has_respiratory, renter_mode?, locale? }</code>	Full replace of the seven booleans — always send all seven . Echoes <code>renter_mode</code> / <code>locale</code> only when sent
<code>DELETE /api/account</code>	<code>{ confirm: "DELETE" }</code>	<code>{ deleted: true }</code> ; 400 if the word is missing or not exactly <code>DELETE</code> . Deletes the account and everything tied to it

D2. Today

`GET /api/daily-score` — no parameters: uses the household's stored location. `?fresh=1` bypasses the one-hour cache (pull to refresh; limited per household, 429 if repeated too quickly). Response:

```
{
  air:    { aqi, severity, sentence, source: "airnow"|"open-meteo"|null, is_measured: true|false|null,
            dominant_pollutant },
            // "PM2.5" | "PM10" | "O3" | null
  uv:    { index, severity, sentence, peak_window: { start, end, max } | null },
            // local "HH:MM"
  pollen: { tree, grass, weed, dominant: "tree"|"grass"|"weed"|null, severity, sentence,
            categories: { tree: { value, severity }, grass: { value, severity }, weed: { value, severity } } },
  mold:  { risk: "low"|"moderate"|"high"|null, is_proxy: true, severity, sentence,
            basis: { humidity_pct, precip_pct } | null },
            // no_data + null risk when the forecast failed
  score: { display_score, score, risk, severity, is_partial, missing_inputs[], included_inputs[],
            includes_proxy, inputs: { air, uv, pollen, mold }, reason },
  cached, retrieved_at
}
```

Errors specific to this route: 400 { error: "No location yet. Finish onboarding, or pass lat and lng." } when the household has no stored location (route to onboarding) · 429 { error: "Readings were just refreshed. Try again in a few minutes.", retry_after_seconds: 300 } when ?fresh=1 is repeated within 5 minutes (keep the readings on screen — A2). lat / lng are still accepted as optional overrides; the frontend doesn't send them.

GET /api/history?days=N (or from/to) → { from, to, count, history: [{ date, score, air, uv, pollen, mold, values: { aqi, uv_index, pollen: { tree, grass, weed }, mold_risk } }] } . One record per local date. Recorded every day for every household by the daily job (not only on days the app is opened), with recent weeks backfilled at onboarding where a historical source exists; a day with no data is simply absent.

D3. HomeGuard

GET /api/home-guard — no parameters: uses the stored county, utility, water source, and build year. Optional county , pwsid , water_source , home_year override them for this request only. Computed on every request — no server cache, no fresh parameter. Error: 400 { error: "No county on file..." } when no county is stored (A2). Water results exist for North Carolina systems only — an out-of-state utility comes back no_data_yet (B3 handles the wording).

Field	Shape
radon	{ county, zone: 1 2 3, risk_level, severity } , or { error } when the county isn't matched, or { error, out_of_state: true, state } for addresses outside North Carolina (radon zones are looked up only when the stored state is NC)
water	Discriminate on water.status: private_well · no_data_yet · lookup_failed · no_pwsid_available · detected_unregulated · anything else = measured. Measured fields: pws_name , population_served , contaminants (every reading: { name: [{ date, date_iso, value_ppt }] }), detected_unregulated_latest[] ({ contaminant, value_ppt, date_iso }), coverage , latest_sample_iso , earliest_sample_iso , data_age_days , reading_count , scored_count , detected_count , detected_unregulated[] , excluded_from_score[] , scored_contaminants[] , includes_guidance , guidance_scored[] , regulatory_notice (only when it applies), message (only when relevant). Private well: source_type , message , test_plan
scored_contaminants[]	{ contaminant, value_ppt, limit_ppt, mcl_ppt, basis, source, is_enforceable, proposed_for_rescission, exceeds_limit, date_iso, risk }
excluded_from_score[]	{ contaminant, value_ppt, date_iso, reason, health_reference_level_ppt, health_reference_source }
test_plan	{ tests: [{ rank, id, name, priority: critical recommended optional, cadence, why, where, estimated_cost: { low, high, is_estimate } }], reasons[], start_here_cost: { low, high, is_estimate }, full_panel_cost: { low, high, is_estimate } }
score	{ display_score, score, risk, severity, is_partial, floored, capped, ceiling, missing_inputs[], included_inputs[], water_risk, radon_risk, reason } . display_score and severity are both null whenever there's no score — a private well or spring, or nothing available (with reason). Same rule for Today's score
lead	{ level, basis, is_estimate, shown_not_scored: true, contributes_to_score: false, text, sentence, utility_inventory } ; null for well/spring. level : high (built before 1950) · elevated (1950-1987) · good (1988 or later) · no_data (year unknown); severe is knowingly unreachable (no address-level inventory exists). utility_inventory : { utility, reported, lead, galvanized_requiring_replacement, non_lead, unknown, needs_replacement, address_level: false, source: { name, url, retrieved } } — the utility's published service-line counts (NC DEQ, July 2025), context only — or null when the utility isn't in the published results
action_plan[]	{ rank, key: water lead radon, severity, title, reason, cost, certification, action } — only items at Elevated or above; text already written for renter or owner
breakdown[]	In order: water, radon, lead (public water only), one per excluded substance. Each: { key, label, severity, contributes_to_score, risk, score, confidence: full limited stale none, is_measured, is_estimate, detail, note } . Excluded-substance boxes add health_reference_level Ug_1 , detail[].value Ug_1

D4. Journal

Endpoint	Request	Response
GET /api/journal	from , to (default last 90 days)	{ from, to, count, entries: [{ id, entry_date, band, symptoms[], severity, note, retrospective, possibly_illness, created_at, updated_at }] }
POST /api/journal	{ entry_date, band, symptoms[], severity, note, retrospective, possibly_illness } — every field, every time (a save rewrites the whole entry). band : one of the seven group keys or household	{ entry } . One entry per date and group; saving again updates
DELETE /api/journal	?id= one entry · ?all=true every entry	{ deleted: n }
POST /api/journal/retrospective	{ dates: ["YYYY-MM-DD", ...] }	{ ready, factor, statement } — compares the flagged days' readings with the rest of that month, e.g. "On the 3 days you flagged, grass pollen averaged High. The month's average was Low." ready: false (with a statement) when too few of the dates have readings
GET /api/journal/findings	—	Not ready: { ready: false, flaggedDays, totalDays, needed: { flagged, total } } . Ready: { ready: true, findings: [{ factor, statement, elevated_rate, other_rate, counts: { elevated_days, other_days } }], disclaimer, flaggedDays, totalDays } (note the camelCase)
GET /api/journal/summary	season , year (default: the season that most recently ended)	{ season, year, logged_days, statement } — e.g. "This spring you logged 22 days. On the 9 highest-pollen days, you flagged symptoms 7 times."

D5. Learn and Assistant

Endpoint	Request	Response
GET /api/learn	topic (pfas · radon · lead · air · pollen · uv · mold), locale , and context: pfas → contaminant, value, limit ; radon → county, zone ; lead → home_year ; air → source, value, severity ; uv / pollen / mold → value, severity	{ topic, locale, title, what_it_is, why_yours, household_note, protect[], sources: [{ label, url, retrieved }] }. locale in the response is the language actually served
GET /api/assistant	page : today · homeguard · journal · map · (none)	{ suggestions: [3 strings], disclaimer }
POST /api/assistant	{ question, page }	Branches — see C2: declined ; configured: false ; grounded { answer, grounded: true, citations: [{ n, label, url, retrieved }], uses_household_data, disclaimer } ; no answer { answer: null, reason: "no_source" "unavailable", message, disclaimer } ; 429 { rateLimited: true, message }

D6. Alerts and notifications

Endpoint	Request	Response
GET /api/alerts	—	{ count, unread, alerts: [{ id, type, severity, title, message, fired_at, read }] }, newest first, dismissed alerts excluded. type : air_quality_change · weather_advisory · new_water_results · radon_season · season_summary
POST /api/alerts	{ id } mark read · { all: true } mark all read · { id, dismissed: true } dismiss · { id, dismissed: false } undo	{ ok: true, updated }
GET /api/notifications · PUT	PUT: { preferences: { air_quality_change, weather_advisory, new_water_results, radon_season, season_summary } }	{ preferences: {...five booleans} }, all default true. The daily job skips types a household turned off
POST /api/push/subscribe · DELETE	POST: the browser's push subscription object (public key: NEXT_PUBLIC_VAPID_PUBLIC_KEY). DELETE: { endpoint }	{ ok: true }. New alerts are also pushed to subscribed browsers with the payload { id, type, title, message, url } (url = the page to open)

D7. Act, Map, District, Sources

Endpoint	Request	Response
GET /api/volunteer	causes (comma list: water, pfas, radon, air, wells, advocacy), county (defaults to the household's)	{ county, causes, count, orgs: [{ id, name, description, counties[], causes[], url, last_verified, match_count }] }
GET /api/map?layer=water	quarter (optional, from quarters)	{ layer, count, located, quarters[], counts: { total, over_limit, hazard_index_over, by_contaminant: { X: { tested, over } } }, features: [{ pwsid, name, lat, lng, population, overall_severity, coverage, latest_sample_iso, detected_unregulated[], hazard_index: { value, severity, proposed_for_rescission } null, contaminants: { PFOA, PFOS, PFHxS, PFNA, "HFPO-DA": { value_ppt, limit_ppt, ratio, severity, date_iso, proposed_for_rescission } { severity: "no_data", limit_ppt } } }], assembled_at }. Systems with only unscoreable detections report overall_severity: "no_data" with coverage: "unscoreable". counts describe exactly the features returned (statewide). quarters uses the form "2024Q3" and is empty when the data has no time series
GET /api/map?layer=radon	—	{ layer, count, features: [{ county, zone, severity }], assembled_at } (county shapes are a static asset bundled with the frontend)
GET /api/map?layer=facilities	—	{ layer, count, features: [{ id, name, lat, lng, programs[], compliance_status, severity, open_violations, recent_violations, penalties, last_inspection }], assembled_at }. compliance_status : no_violation · violation · significant_violation · unknown . recent_violations = count over the last 3 years; penalties = dollars over the last 5 years
GET /api/map?layer=air	—	{ layer, count, features: [{ id, name, lat, lng, aqi, severity, dominant_pollutant, last_reading_at }], assembled_at } (dominant_pollutant as in D2)
GET /api/district	—	{ scope: "NC-08", total_systems, systems_over_limit, affected_population, affected_population_complete, by_contaminant: { X: { tested, over, limit, population_over } }, exceedance_lines: { X: string }, counties: [{ county, systems, systems_over_limit, worst_water_severity, radon_zone, radon_severity }], state_comparison: { district_share_over, state_share_over }, household_county: { county, water_rank, of }, assembled_at }. Exceedance lines count only systems tested for that compound. Everything except household_county is NC-08-wide and pre-built daily; household_county is filled in per request from the caller's

		profile
GET /api/sources	—	{ sources: [{ name, provides, last_retrieved }] } — Settings → About and the health card's sources footer

Map and district data are built once a day by the scheduled job and served from storage, so they load fast and can't break during a demo because a government service is slow (\$13.10, \$62.1).

Part E — Definition of Done and Build Order

E1. A page is done when LOCKED

Not when it renders (§42). Every box, every page:

☐	All four states — loading, error, empty, populated — with different copy for error and empty
☐	Every value formatted per A5; every string from the content inventory or approved as PROPOSED — no invented copy
☐	Every severity shown as color + icon + word; missing data as an em dash or an explicit state, never zero or green
☐	Every estimate labeled as an estimate; every data point's source and date available to the person
☐	Bound to the exact fields in Part D; nothing computed that the server decides; no legacy <code>status</code> fields
☐	Every field on the page tested absent/null without breaking
☐	Back button correct for any sheet it opens; scroll position kept on return
☐	Keyboard navigable with visible focus; sheets trap and return focus; 44px targets; reduced motion respected
☐	Checked at 375px on a real phone, in light and dark mode
☐	No console errors; meets the performance targets (first content under 1.5s, cached readings under 200ms, tab change under 100ms, map first render under 2s)
☐	Every string goes through the translation layer; both languages present once reviewed Spanish exists
☐	(Once, app-wide) Page titles, link-preview metadata, <code>theme-color</code> , favicon and home-screen icons set per A5's identity rules

E2. Build order

#	Build	Why here
0	Design foundation: severity colors and icons, brand palette, type (locked — see A11)	Every page inherits it
1	Shared layer: data hook + state wrapper, auth handling (401 vs 503), formatting module, severity mapping, translation layer, color tokens (light + dark)	Parts A2–A5; cheap now, a rewrite later
2	Onboarding (B1) + results reveal	The entry point; sets button, input, chip, and spacing styles; lowest-risk place to validate the design foundation
3	Shell: header, tab bar, routing, back contract, assistant button	Everything renders inside it
4	Today (B2) — on a real phone, with live data	The blueprint's own measurement (§26.4): if this takes hours, the plan holds; if days, cut scope now
5	HomeGuard (B3) + Learn sheet (C1)	First bottom sheet; the deepest data view
6	Journal (B4)	
7	Act (B5) + Alerts sheet (C3)	Mostly composes data already loaded
8	Settings (B6)	
9	Map + district panel (B7), then Assistant sheet (C2)	Largest builds; the blueprint's own cut order if time runs short is Assistant → Settings → Act → district panel

Two finished pages beat five half-finished ones under every judging criterion, and an unfinished page reads as broken rather than scoped (§26.3).

Part F — Decision Log

Every decision that shapes the frontend, made during planning (September 2026). **LOCKED** = decided, build to it. **PROPOSED** = a recommendation awaiting Yogi's confirmation. A superseded decision is listed with what replaced it, so an old document or branch can't reintroduce it.

F1. Product and scope

#	Decision	Status	Why / notes
1	HALO is a web app with a mobile-shaped layout, running in a phone browser — not a native iOS or Android app	LOCKED	§3. The Capacitor config in the repo isn't part of this build
2	The blueprint's function, content, buttons, states, copy, and data rules are ground truth. Its visual and style spec is void and redesigned	LOCKED	Team decision. This document carries the function; A11 and A13 carry the new look
3	Vibhav's glassmorphic HomeGuard frontend (branch <code>vmag211/homeguard-frontend</code>) and its <code>docs/HALO-design-language.md</code> are discarded	LOCKED	With Vibhav's permission. Don't merge, copy, or follow them
4	Every page holds all the content it can; Onboarding is the deliberate exception and stays minimal	LOCKED	§10.1 — every extra onboarding question costs completions
5	This specification targets the finished backend; Part D is the contract both sides build to	LOCKED	The absent-field rule (D0) covers anything not live yet
6	Build order: foundation → shared layer → Onboarding → shell → Today → HomeGuard + Learn → Journal → Act + Alerts → Settings → Map + district → Assistant	LOCKED	E2. Cut order from the end: Assistant, Settings, Act, district panel
7	Codex does all frontend work — wireframes, visual design, and code. Yogi supplies reference inspiration and approves each page	LOCKED	September 27. Process in A13
8	Spanish ships only after a person reviews it; every string goes through a translation layer from day one	LOCKED	§68 — machine-translated health guidance is not acceptable

F2. Design

#	Decision	Status	Why / notes
9	Six severity words — good, moderate, elevated, high, severe, no_data — exactly as the backend's <code>lib/severity.js</code> emits them	LOCKED	An earlier idea to drop "elevated" was superseded: the scale follows the existing code
10	Severity colors: Good green, Moderate orange, Elevated red, High a red-leaning warning purple, Severe a darker purple, No data black drawn as an outline only	LOCKED	Exact light and dark values in A11. The palette wasn't tuned for red-green color blindness — the team's choice — which is safe only because severity is always color + icon + word
11	Brand: dark blues, navy, dark teals, turquoise, with a sprinkle of light aqua; no saturated green in the brand	LOCKED	A11. Green always means Good
12	Typefaces — the "field journal" set: Fraunces (headings, hero scores), Instrument Sans (everything else, tabular digits for data), IBM Plex Mono (codes only)	LOCKED	Chosen over more generic candidates for personality without losing clarity
13	Light mode is the default; dark mode follows the system setting; no in-app toggle	LOCKED	§59. Every color is a token from day one
14	The palette swatch sheet and type specimen made during planning were previews of colors and fonts only — not page designs	LOCKED	Page design happens per page, in Codex
15	Onboarding tone: "something in between" — light and calm, with a navy-to-teal brand band behind the logo on Welcome only	LOCKED	Details in B1
16	Base visual system — type scale, spacing, radius, buttons, inputs, chips, progress, pills, motion	PROPOSED	A13. Finalized in Onboarding's design pass
17	The blueprint's simplifications are the defaults: three button variants, two sheet heights, pill rather than colored card edge, two skeleton shapes, three motion durations	LOCKED	§65.4 (A9 note)

F3. Data, privacy, and backend behavior the frontend depends on

#	Decision	Status	Why / notes
18	The server decides every severity, score, and household sentence; the frontend never applies a threshold	LOCKED	§6.1. Bind <code>severity</code> , never the legacy <code>status</code>
19	401 = session definitively invalid → reset silently. 503 = couldn't verify now → retry, never reset	LOCKED	Protects a household's data during an auth outage. Backend fix is on <code>vmag211/backend</code>
20	Route on <code>onboarding_complete</code> (not <code>onboarded</code>), backed by the local <code>halo.onboarded</code> flag	LOCKED	A1
21	The Assistant may send the relevant readings and journal symptom names and dates to its AI provider — never the free-text notes. The privacy statement says so	LOCKED	September 27. Acceptable because HALO isn't shipping as a published product. Statement text in B6

22	Stored coordinates are rounded to about 100 m; ZIP is no longer stored	LOCKED	Matches the privacy statement's "approximate coordinates"
23	Radon zones apply only to North Carolina addresses; others get <code>out_of_state</code>	LOCKED	County names repeat across states (Union County, SC)
24	The Severe lead level is knowingly unreachable — no address-level service-line inventory exists; the utility's published counts are shown as context	LOCKED	Recorded by the backend, September 27
25	Journal's first-visit comparison is computed server-side (<code>POST /api/journal/retrospective</code>)	LOCKED	The frontend never compares readings itself
26	Map layers and district figures are pre-built daily and served from storage	LOCKED	Fast, and a slow government service can't break a demo
27	Anonymous sign-in is enabled in the Supabase project	LOCKED	Confirmed by Yogi. Re-check if the project is ever recreated (§66 item 1)
28	Settings toasts only on failure (the blueprint contradicts itself: §49.2 vs §54)	PROPOSED	Quieter; one rule for all of Settings

F4. Still open

- Every **PROPOSED** item in Parts A–C — Yogi keeps or cuts each as its page is built.
- The logo's treatment on Welcome's navy band (B1).
- Where the Settings "Change address" flow ends (B6 proposes: address → Your home → results reveal → Today).
- The alerts badge: clear as items are read (proposed) or on opening the sheet (§54) (B2).
- The privacy statement's incomplete "stores" line — a proposed addition awaits approval (B6).
- With Vibhav: server timeouts per source (§62.2 targets aren't enforced), reloading water data with non-detects (the map's "no detection" label), a distinct water status for out-of-state utilities, and invalidating the daily cache when the address changes.
- The map library — Mapbox GL JS on the existing Mapbox account (needs a public, URL-restricted token) or MapLibre with free tiles — and the source and size of the bundled NC county shapes (B7).
- Whether the map time slider exists — depends on whether the stored water data has multiple dated readings per compound (§66 item 5, §68). The loader kept every detected reading with its collection date, so the answer is probably yes; one database query settles it. If not, `quarters` is empty and the slider isn't built.
- Whether Spanish ships — depends on a reviewer (§68).
- Whether the Assistant ships — it's last in the build order (§68).
- Which evaluation criteria the NC-08 office uses this year — one email to ask (§66 item 12).